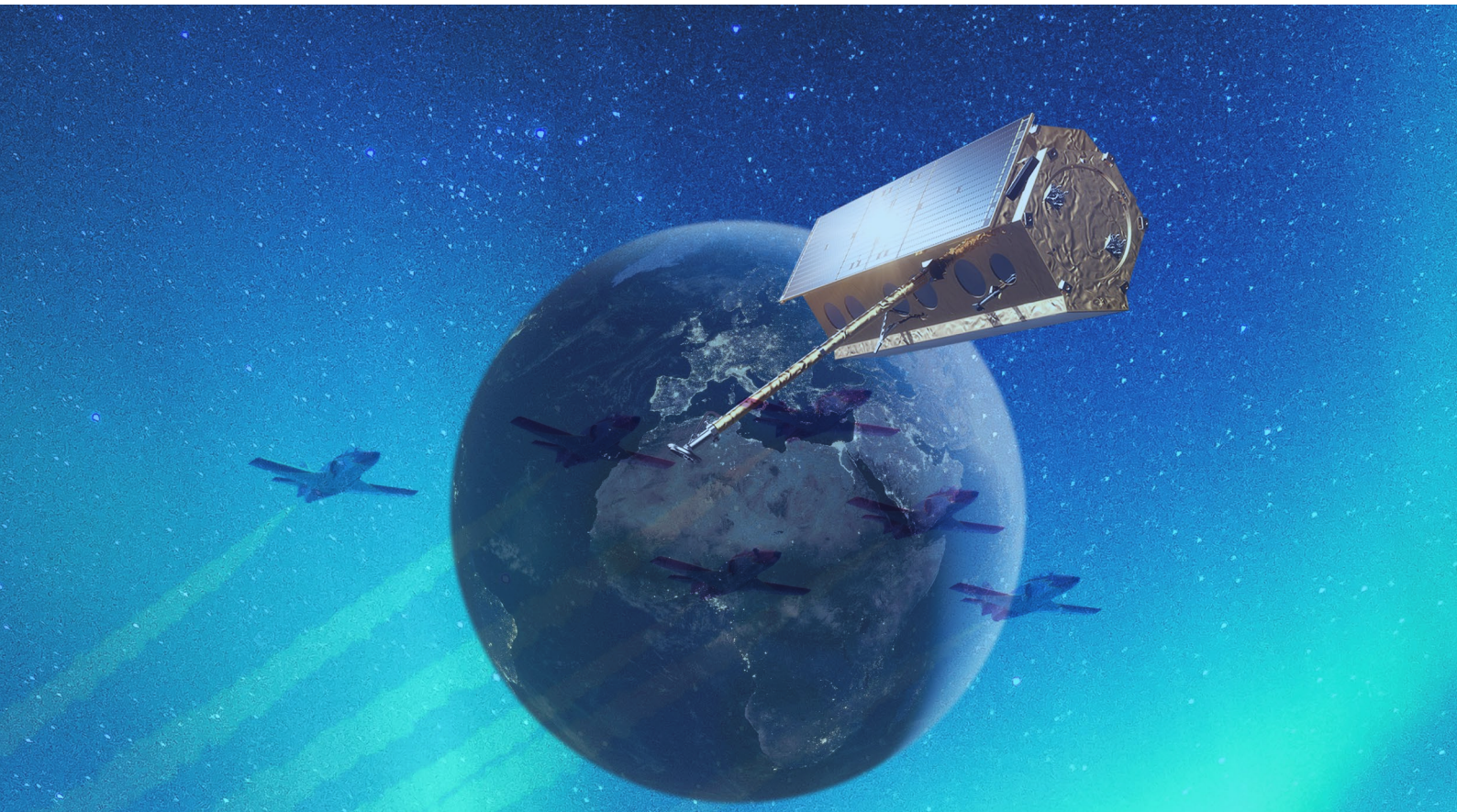
NATIONAL AEROSPACE SECURITY STRATEGY 2025



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NATIONAL AEROSPACE SECURITY STRATEGY

2025

The National Aerospace Security Strategy was approved by the National Security Council on 14 July 2025.

The following entities participated in the development process: the Ministry of Foreign Affairs, European Union and Cooperation; the Ministry of the Presidency, Justice and Parliamentary Relations; the Ministry of Defence; the Ministry of Finance; the Ministry of Home Affairs; the Ministry of Transport and Sustainable Mobility; the Ministry of Labour and Social Economy; the Ministry of Industry and Tourism; the Ministry for Ecological Transition and Demographic Challenge; the Ministry of Economy, Trade and Business; the Ministry of Health; the Ministry of Science, Innovation and Universities; the Ministry for Digital Transformation and Public Function; the Spanish Space Agency; the National Intelligence Centre; and the Department of National Security.

LETTER FROM THE PRESIDENT OF THE GOVERNMENT

The nature of conflicts has evolved. We are witnessing this transformation in Ukraine and the Middle East, as well as in many other regions across the globe. Consequently, we must be prepared to confront the new threats that lie ahead.

One of the areas where these changes are most evident is aerospace. Here, emerging geopolitical dynamics, the rapid adoption of advanced technologies, and pressing environmental challenges define a new context—one that compels us to look toward the future with ambition and fresh perspectives.

By its very nature, the aerospace sector is dual-use, serving both civil and military purposes. In our daily lives, numerous technologies originally developed for military applications—such as radar, navigation systems, and satellite communications—are indispensable. The reverse is also true: civil and military aeronautical and space industries are deeply interconnected. For this reason, security cannot be separated from industrial policy.

Security in the aerospace domain is equally critical for sectors vital to Spain's economy, such as tourism. This concept of security must be understood broadly: encompassing deterrence and air defense capabilities, health protection, cyber and supply chain security, and responses to threats such as terrorism and organized crime. Furthermore, training and attracting talent in aerospace disciplines is an essential component of our security framework.

LETTER FROM THE PRESIDENT OF THE GOVERNMENT

Spain must position itself as a nation committed to air and space, embracing aerospace development from a security perspective. In today's context of uncertainty and intensifying geostrategic competition, the 2025 National Aerospace Security Strategy provides high-level guidance to foster collaboration among all key stakeholders—ranging from public administrations and the aerospace sector to our international allies.



Pedro Sánchez

President of the Government of Spain

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EXECUTIVE SUMMARY



Figure 1: Spanish Space Agency Headquarters in Seville. Image: AEE

EXECUTIVE SUMMARY

Spain and Europe as a whole are intensely affected by increasing global geopolitical instability. The international order is undergoing changes of a magnitude not seen since 1945, and we cannot remain passive onlookers before the new global order that is rapidly taking shape. Instead, we must actively participate in constructing the emerging international structure. But to play a decisive role in shaping the new international order to reflect our own values and interests amid growing competition among powers of different stripes, we must strengthen our capabilities and reduce strategic dependencies in order to mitigate significant vulnerabilities that currently affect our security.

Russia's war of aggression against Ukraine has starkly exposed our shortcomings and vulnerabilities. Only through joint efforts to strengthen our capabilities will we reduce the severe dependencies that are now preventing us from making independent decisions and which, ultimately, could jeopardise our future.

The aerospace sector is one of the areas that most clearly reveals these vulnerabilities and dependencies. The White Paper on European Defence, presented by the European Commission on 19 March 2025, at the request of the European Council, articulates a compact and coherent strategy that includes joint, ambitious and necessary measures to address these deficiencies over the next five years. This is a true emergency plan, one that will mitigate decades-long shortcomings if its proposals and approaches are implemented by us all, working together.

The 2025 National Aerospace Security Strategy (NASS) is the second national strategy in this area, succeeding the first of its kind, approved in 2019. Since then, rapid and significant changes in the air and space domains have had a direct impact on the aerospace sector, which is of vital importance to national security. This dynamic of change underscores the need to review the 2019 NASS, as well as the aerospace-related provisions included in the 2021 National Security Strategy (NSS), with a view to expanding existing provisions and to incorporating new aspects related to aerospace security, within and beyond the atmosphere.

In this regard, the 2021 NSS proposed a line of action to ensure the security of global common spaces, via the creation of a Spanish Space Agency. This line of action was formally defined by Royal Decree 158/2023, which approved the Statute for the Spanish Space Agency, with a component dedicated to national security. Moreover, Line of Action No. 3 of the 2021 NSS recommended "*Developing the industrial complex for defence, security and space, and developing dual technologies...*" The importance of this focus is highlighted by the significant recent shift toward industrial policies aimed at ensuring greater economic security at the international level. Therefore, it is necessary to define new strategic lines of action in the field of aerospace security.

The Industrial and Technological Plan for Security and Defence, approved by the Council of Ministers in April 2025, is Spain's response to the current international context, within the Government's firm commitment to the security of its citizens, to the country's technological progress and to furthering the European project.

Like its 2019 predecessor, the 2025 NASS considers it advisable to view aerospace security as a single question, both to effectively address threats and to mitigate common and interdependent risks within and beyond the terrestrial airspace.

However, in a context of increasing competition at all levels, this integrated approach requires us to recognise, simultaneously, that outer space represents a distinct domain, where threats to services essential to national security have intensified. Both the European Union and NATO have denominated outer space as a distinct operational domain, while implementing a comprehensive and integrative approach to terrestrial airspace and outer space to ensure the effective coordination and management of resources.

The Strategy is divided into four chapters. The first, entitled " Contextual analysis of the air and space domains", presents general aspects, briefly describes the regulatory framework and sets out the need for a single aerospace security strategy to protect key national interests in both these domains.

The second chapter, entitled "Threats to National Security in the Domains of Air and Space," addresses the concepts of risk and threat in the field of aerospace. It identifies the main vulnerabilities which arise from the nature of these domains: their level of interconnectivity, the highly technological nature of the operational infrastructure, the dependence on industrial capabilities, and the reality of constant technological advances.

Due to the unique characteristics of the aerospace sector, many of the threats that may arise lie outside Spanish sovereignty and jurisdiction, making it necessary at all times to consider both national responsibilities and Spain's international commitments.

The third chapter, "Objectives," details the main strategic targets for the 2021 NSS, setting out the following objectives and lines of action for the aerospace sector.

- **Strategic Objective 1:** Protect Spain in and from air and space, with the fundamental understanding that Spain should "PROTECT people's lives, rights and freedoms, and defend the Constitutional order".

- **Strategic Objective 2:** Improve security and defence capabilities in the air and in space, with the understanding that Spain should "PROMOTE the prosperity and well-being of its citizens."

- **Strategic Objective 3:** Make air and space safer and more sustainable, with the understanding that Spain should "PARTICIPATE in the preservation of international peace and security and defend its strategic interests".

Each of these strategic objectives is associated with various lines of action to provide strategic political guidance to the relevant ministerial departments, which will decide on the programmes to be implemented within their respective areas of responsibility.

The fourth chapter, "Aerospace Security in the National Security System," reviews the organic architecture of aerospace security as previously defined in the 2019 NASS. Headed by the President of the Government, this structure is composed of three permanent bodies (the National Security Council, the National Aerospace Security Council and the Situation Committee). In addition, a Coordination Group may be activated.

A new feature of the 2025 Strategy is the inclusion of an Annex, detailing specific measures aimed at achieving the three Strategic Objectives. As with the lines of action, these measures will contribute to achieving more than one objective. However, to simplify monitoring their performance, each measure is classified under a single Strategic Objective.



Figure 2: Radio Navigation Facility, Santiago de Compostela Airport. Image: MITMS-ENAIRE



INTRODUCTION

INTRODUCTION

In 2019, the National Security Council (CSN) approved the first National Aerospace Security Strategy (NASS). Among its main innovations was the proposal to create the National Aerospace Security Council (CNSA), modelled after similar bodies established in other areas of national security. Supporting the CSN, the CNSA evaluates the progress and implementation of the NASS.

In the course of its work, the CNSA identified the need to update the NASS. This proposal was approved by the CSN, for the following reasons.

First, the global strategic context is undergoing significant geopolitical transformations, characterised by a notable increase in regional tensions and conflicts. In this respect, the current situation highlights the great difficulty in ensuring the security of terrestrial airspace and outer space, given the emergence of new threats to space capabilities and the proliferation of vectors that, although mentioned in the 2019 NASS, have since undergone significant expansion and technological evolution.

In the case of drones, there has been exponential growth, both quantitatively and qualitatively, in their use in military and terrorist operations, driven by the lethal use of small commercial and military drones, as demonstrated in recent conflicts (Ukraine, Gaza, the Middle East, etc.). Furthermore, there has been an emergence and multiplication of the use of loitering munitions, which are increasingly accessible and have even more lethal effects. Likewise, there has been a significant development of ballistic, cruise and hypersonic missiles, as well as long-range precision weapons launched from the ground or from the air.

Another disturbing phenomenon is the proliferation of incidents at airports, with the sighting of commercial drones in their vicinity. The consequent danger of interference with navigation systems or even with the aircraft themselves has sometimes forced local air traffic to be diverted, for operational safety.

The second outstanding question, and motive for reviewing the former Strategy, is the need to constantly assess the threats evolving in previously unknown spheres. The emergence of new players in the field of aerospace, some of them beyond the control of States and of traditional international structures, is not unheard of, but the rapid increase in their number and in their technological and financial capabilities is novel and alarming.

The evolution of these and other threats requires concerted effort to ensure cybersecurity and the protection of critical entities within the aerospace sector. It is essential to maintain the products, applications and services provided by space capabilities such as communications satellites, Earth observation satellites, meteorological satellites, and positioning, navigation and synchronisation systems.

Thirdly, within the framework of the Strategic Compass objectives, which aim to strengthen the European Union as a security provider, in March 2023 the EU Commission and the High Representative presented the Joint Communication "European Union Space Strategy for Security and Defence" to the European Parliament and the Council. Another initiative, the White Paper on European Defence, identified drone/counter-drone systems and air and missile defence as priority capability areas. The

2025 NASS offers the opportunity to determine national priorities and to influence the development of the EU strategy in alignment with these priorities.

The necessary parallelism between European and national policies on aerospace security and defence is a factor that generates synergies and necessitates a synchronised update of both strategies. The present review of the Strategy emphasises not only its national character, but also the need to strengthen cooperation with other States and organisations with which Spain shares principles and political systems, as well as technological, economic, financial and commercial capabilities.

Finally, the present moment is highly appropriate, considering the evolution of the national aerospace regulatory and jurisdictional sphere, with the creation of the Spanish Space Agency and the adaptation of the Ministry of Defence, with the change of name from the Spanish Air Force to the Spanish Air and Space Force and the creation of the Space Command, to implement new competencies in outer space.

Under Spanish legislation, Article 3 of Act 36/2015 defines national security as " *action taken by the State to protect the freedom, rights and well-being of citizens, to ensure the defence of Spain and its constitutional principles and values, and to contribute, together with our partners and allies, to international security in fulfilling the obligations undertaken.*" With respect to the context of aerospace, it should be taken into account that the pre-existing regulatory framework, both national and international, often uses the generic term "security".

The NASS, based on the concepts used by the International Civil Aviation Organisation (ICAO), uses the following definitions of safety and security, terms that are related but not interchangeable.

- *Safety*: A state in which the risks associated with activities related to the operation of aircraft, or directly supporting such operation, are reduced and controlled to an acceptable level.

In this definition, the risk to operational safety should be understood as the expected probability and severity of the consequences or results of a hazard, i.e., the condition or object that could cause or contribute to an aviation incident or accident.

In other words, measures aimed at operational safety seek to proactively mitigate operational safety risks before they result in aviation accidents or incidents.

- *Security*: A condition in which risks and threats that may compromise the State's freedom of action in its sovereign airspace, or where it bears responsibility, and those that may affect the safety of citizens using the air domain, are reduced and controlled to an acceptable level. It includes the following elements:

  - Surveillance of outer space, surveillance and control of airspace, and intervention when necessary.

  - Defence against hostile or potentially hostile actions within the air domain.

  - Contributing to air safety and defence both domestically and abroad, within the framework of the international organisations to which Spain belongs.

  - Combating the communication of false information that might compromise the operational safety of an aircraft, whether in flight or on the ground, or

the safety of passengers, crew, ground personnel and/or the public at an airport or on the premises of a civil aviation facility.

Although these concepts are distinct, the aviation community recognises that many risks concerning safety, security or other areas like health security (such as the COVID-19 pandemic) cannot be addressed in isolation due to their interconnectedness and cross-cutting impact. Therefore, collaboration across different areas is essential.

An example of the need to jointly manage operational safety and security is the case of higher airspace operations (HAO), which have been provisionally defined as those conducted above the altitudes at which aviation services are currently provided (up to approximately 55,000 feet). High-altitude pseudo-satellites, remotely piloted vehicles operating at high altitudes, balloons, and supersonic or hypersonic aircraft or missiles can all operate at these altitudes, crewed or otherwise, autonomously or with varying degrees of automation. High-altitude operations pose risks to operational safety (to other air traffic, third parties, or persons and property on the ground), but the risks to security and defence are no less significant.

Above these altitudes, it is difficult to exercise sovereignty due to the limitations of surveillance and control systems and of air defence operations. Detection by conventional radar, whether ground or aircraft-based, is difficult at these altitudes, and the high speed of aircraft can make interception difficult. However, if left unchecked, HAO by external forces pose a real threat.

In view of the particularities of the space domain with respect to the air domain, the ICAO-defined terms must be adapted to maintain their functionality and relevance.

In accordance with the definitions proposed by the United Nations Office for Outer Space Affairs (UNOOSA), the concept of "safety" could be applied to operational safety in space, viewed as the result of measures aimed at preventing accidental or unintentional damage that impedes normal operation and at mitigating potential damage to space assets. On the other hand, "security" would refer to protection against threats to all aspects of space systems, in order to prevent deliberate damage.

This interpretation is applied in the Statute of the Spanish Space Agency, which is responsible for supporting the supervision of both safety and security in this sphere, in coordination with the competent national agencies. In addition, the Spanish Armed Forces (FAS), through the Space Command and the Spanish Air and Space Force, have jurisdiction in the areas of security and defence, particularly with respect to space systems of government interest.



Figure 3: HISDESAT Satellite Control Centre. Image: HISDESAT



CHAPTER 1

CONTEXTUAL ANALYSIS OF THE AIR AND SPACE DOMAINS

GENERAL ASPECTS OF AEROSPACE AND RELATED ACTIVITIES

The development of commercial aviation, the important economic activity generated within airspace and in its use, and the harmful effects on our society that can result from the control and utilisation of this environment by antagonists, even in peacetime, make the domains of air and space a fundamental area of State action.

The National Security Act of 2015 includes the security of terrestrial airspace and outer space among the areas of special interest for national security that require specific attention. These spaces have no physical boundaries and are characterised, among other parameters, by their accessibility. In the case of outer space, this is facilitated by the increasing use of low-altitude orbits.

Outer space, henceforth simply "space," is a continuous global environment without geographical barriers or national sovereignty, governed by its own physical laws and only loosely regulated by international treaties. It hosts a growing number of capabilities and services crucial to society, security and defence, which in turn has intensified the presence and activity of both State and non-State actors.

The environmental complexity of space, together with the diverse interests of and relationships among the parties involved, highlights the need for appropriate management and regulation. Although some critical aspects of space use, such as the assignment of orbits and operating frequencies, are rules governed, it is imperative to develop a more detailed framework that effectively regulates the safe and sustainable use of space.

Space, particularly Earth orbit, is a competitive, contested and congested environment, requiring specific capabilities to guarantee safe and secure access and thus contribute to decision-making autonomy. These circumstances have led Spain, the European Union and NATO to recognise its importance as a specific operational domain for security and defence.

Security in Aerospace (terrestrial and extra-terrestrial space) must be addressed through coordinated policies, measures and actions to protect and ensure freedom of action consistent with the proper use, integrity, functionality and sustainability of both domains.

Spain, given its capabilities and economic position, is one of Europe's main partners in the aeronautical and space sectors. Each of these is crucial to the nation's economic development and security, and consequently, it is essential to guarantee access and freedom of action in both areas.

Aerospace assets are considered critical entities and essential services as established by the European directives on the resilience of critical entities (CER) and on cybersecurity (NIS2). The protection of these and associated infrastructures, as well as the proper management and safeguarding of the integrity of capabilities, users and services, is essential for the functioning and development of society.

The lack of a physical boundary between the air and space domains makes international agreement on its definition difficult; hence, existing regulations on the matter are limited and imprecise. For the purposes of the Strategy, and to facilitate the coordination of operations in both domains, in this paper the theoretical and academic boundary between the two domains is considered to be located at an altitude of approximately 100 km, on the so-called Kármán line.

The rules governing the use of air and space are based on substantially different principles. While national airspace is subject to State sovereignty, outer space is considered the common heritage of humanity, according to international treaties. However, the principle of non-appropriation of space, enshrined in United Nations treaties on the matter, is subject to interpretation and could be challenged in the future by national laws that facilitate the commercial exploration and use of space resources.

The exploration and exploitation of space represents a source of knowledge, drives the development of new technologies, provides applications that have become indispensable for quality of life in society, creates highly qualified jobs, and has significant implications for security and defence. The demand for services related to the use of space will continue to increase. Accordingly, the State must support the sector, seeking the greatest possible strategic autonomy, and reducing its technological and industrial dependence on other powers in the development of space capabilities and the provision of services essential to the functioning and economy of the country.

The importance of airspace is not limited to the rapid and efficient transport of people and goods; this domain is also fundamental to global integration, contributing to economic growth and connectivity between nations. Like space, it is a strategic sector that generates highly skilled employment and vital investment in technology and innovation, and is a fundamental driver of exports.

The growing demand for aeronautical services requires us to reduce their environmental impact and improve operational efficiency. In consequence, State support is needed to encourage the development of proprietary technologies and to reduce our external dependence for critical components.

PATTERNS OF CHANGE IN THE DOMAINS OF AIR AND SPACE

The international context is undergoing a period of great uncertainty, although certain clear trends are emerging that will shape the global order in the medium to long term. This combination of changes and constants is shaping a new paradigm in political, economic and geopolitical spheres.

FACTORS OF UNCERTAINTY

Two factors make it difficult to contemplate the present time in terms of traditional patterns of competition among States:

- **Challenges of a global nature** – primarily climate change, but also others such as pandemics, certain technological aspects or nuclear proliferation – can only be addressed through international cooperation, which is more difficult in a context of rivalries. These foster mistrust, which incentivises States not to cede relative advantages, even if the price of non-cooperation is higher.

- **The accelerating adoption of new technologies**, the consequences of which are unpredictable. The aerospace sector is just one arena within this transformation, but it is interconnected with many others. The weakening of our industrial and technological base may lead to difficult choices between acquiring capabilities (created by foreign actors, public or private) or maintaining higher levels of autonomy and sovereignty.

The digital revolution – with all its corollaries, including the collectivisation of knowledge through the Internet, its automatic replication via artificial intelligence (AI), the blurring of the virtual and physical realms, quantum computing, mass data collection and its behavioural predictive power, the manipulation of information and images, and the automation of warfare – impacts all areas of international relations and is a factor of uncertainty.

Technology has become a source of power, rivalry and competition, in many spheres of life, including the military. It has transformed the fabric of the global economy, its infrastructure, and the very concepts of value creation, fostering the globalisation of its services (led by the new technology giants) and highlighting the need for regulatory harmonisation, a goal that conflicts with growing geopolitical fragmentation.

NEW CONSTANTS

There is a growing use of force outside the scope of international law, the influence of the Indo-Pacific is on the rise, and Europe is facing increasing conflicts. At the same time, multilateralism is weakening and the Global South is gaining economic and demographic prominence, albeit with a worrying rise in transactional diplomacy and a downplaying of international law and human rights.

In the economic sphere, globalisation has heightened concerns for resilience and economic security. Risk mitigation is being prioritised over freedom to trade, creating barriers and disputes in global commerce. Businesses can no longer isolate themselves from international politics, and are becoming more present in States' foreign policies.

Security is no longer dissociable from industrial policy. We are witnessing a shift in mentality among some of the world's leading industrial powers, who treat the competitiveness of their companies as a strategic requirement as well as an economic asset. This new mentality poses serious dilemmas for economic models like those of Spain, and Europe as a whole, which are based on open trade, free markets and free competition.

In line with these underlying geopolitical trends, there is growing competition in space among countries such as China, Russia, India and the United States, which are investing heavily in advanced space technologies.

The above are some of the underlying trends that, apart from turning points like the war in Ukraine and future unforeseeable events, are decisively transforming the world and intertwining geopolitics and economics to an increasing extent.

PATTERNS OF CHANGE

From an economic perspective, the aeronautical industry is an economic mainstay, generating employment and contributing massively to our GDP. The sector's competitiveness must be maintained through investment in innovation and improvements throughout the value chain.

To support the sector's growth and improve safety, airports must be modernised and advanced air traffic management systems developed. Moreover, such infrastructure investments boost tourism and trade.

Governments in all major developed countries provide financial incentives and support to their domestic industries to encourage innovation and protect employment in the aeronautical sector.

The space sector, too, is a driver of growth and competitiveness. However, it requires greater investment in advanced technologies and more support for public-private collaboration. Furthermore, rapid industrial and technological development in the space domain offers new market opportunities and can further the creation of highly skilled jobs in high-tech sectors, positioning Spain as a significant player in this context. Finally, the space domain is of crucial importance in monitoring and mitigating climate change, offering significant opportunities for the development of a green and sustainable economy.

In the technological dimension, the contest is particularly intense, with many countries jockeying for global supremacy, often controlling exports of critical and dual-use technologies. The geopolitical importance of the aerospace sector is reflected in the exponential increase in new strategic defence programmes in this area (for example, to develop hypersonic missiles, AI, directed energy and drones). EU policymakers emphasise the need to ensure the Community's autonomy and sovereignty in strategic technologies through programmes such as the European Defence Fund (EDF).

Innovation in the aeronautical sector requires a long timeframe and significant financial investment in R&D, whilst maintaining the contribution to reducing emissions

and providing functionality, as each component must comply with strict European and national aviation safety standards. Therefore, institutional support in this area, through specific national R&D programmes, is essential.

With an eye to technological advance and the quickening introduction of more competitive technological developments, a relevant consideration is the development of new capabilities related to security and defence, to protect space systems of national interest and to ensure freedom of action in space.

There are numerous synergies between disruptive technologies that have been developed, promoted and/or applied to space systems and other areas of national technological development, with manifold applications, such as robotisation, AI, optical communications, quantum computing, biotechnology, energy generation/storage and additive manufacturing.

In the environmental dimension, Spain has positioned itself as a key player, committed to the transition toward a more sustainable and environmentally friendly future, and to harnessing the potential of space technologies and programmes. In line with the UN 2030 Agenda and its 17 Sustainable Development Goals (SDGs), Spain is resolved to exploit the potential of space to address global challenges such as climate change, the management of natural resources and the need to protect our planet.

Space technology contributes significantly to the SDGs, for example by enabling Earth observation, helping optimise agriculture and natural resource management, improving disease surveillance and disaster response, and enhancing connectivity and distance learning. It also facilitates the integration of renewable energy and climate change mitigation and, via geospatial data, supports urban planning and informed decision-making.

Spain is firmly committed to help achieve international agreements for environmental protection, particularly those regarding climate change, such as the Paris Agreement and the EU's goals in this respect (including the Fit for 55 package and the European Green Deal).

In the field of air transport, Spain is a strong supporter of the International Civil Aviation Organization (ICAO) Global Framework, which establishes an initial milestone of a 5% net emissions reduction worldwide by 2030, thanks solely to the use of cleaner energy sources. Moreover, EU-wide policies are being actively promoted to achieve climate neutrality in air transport.

Spain has the opportunity to become a leader in sustainable aviation fuel (SAF) production, a position it already holds in that of other renewable energies. Since Spain's SAF production potential far exceeds local demand, the installation of more production plants would allow large volumes of this innovative fuel to be exported to the international market, with the resulting environmental and, at the same time, economic benefits.



Figure 4: Mérida SAR Image. Image: HISDESAT

SPAIN, IN THE AIR AND IN SPACE

Spain's privileged geographical location, its natural surroundings and unique cultural environment have consolidated its position as one of the world's leading tourist destinations. In 2024, a record 93.8 million tourists visited Spain, a 10.1% increase over the previous year. Accordingly, aerospace safety, in all its dimensions, must be addressed to ensure these travellers are ensured all possible safety.

ENAIRE is the leading provider of air navigation and aeronautical information services in Spain, the fourth largest in Europe by traffic volume, and one of the largest worldwide. AENA manages 46 airports and two heliports in Spain, and through its subsidiary AENA International operates 18 airports in the United Kingdom and Brazil and participates in the management of another 15 airports in the Americas. The proven and continuing success of the integrated and networked management models of ENAIRE and AENA is a vital part of the development of the airspace.

The Spanish aerospace industry participates in all major European aeronautical developments, both civil and military. Spanish companies excel in the provision of military transport aircraft, turbines, maintenance services, and components for space missions, air traffic control centres and detection/tracking systems (radars), and have gained a solid reputation and widespread trust. Moreover, Spain is one of the few countries with the capacity to cover the entire aircraft design and manufacturing cycle.

The Spanish space sector has grown significantly over the last five years. The capabilities provided by leading medium- and large-scale companies have served as a driving force for an entire ecosystem of small and medium-sized enterprises that contribute their expertise to the development of this sector. Public-private collaboration models, increased funding opportunities and the creation of specialised industrial hubs, among other factors, have been decisive in the development and consolidation of a space industrial network that offers and exports innovative solutions.

In the European context, Spain has been a member of the European Space Agency (ESA) since its foundation in 1975 and participates in its main programmes. It is also an active partner in the EU Space Programme, which promotes and supports innovation

and competitiveness through investments in disruptive technologies. The components of the EU Space Programme should also be considered of interest in terms of national security, due to their strategic nature, but they do not specifically address defence needs.

The creation of the Spanish Space Agency (AEE) has boosted space research and strengthened the national industry in this sector, providing Spain with a solid technological and industrial base that, aligned with national security interests, has heightened our technological independence and made us better placed to face new challenges and opportunities within a context of international cooperation.

The existence of a consolidated aerospace sector, fundamental to our economy, together with the development of strategic national programmes, participation in European programmes of great technological and industrial interest, a commitment to innovation, the significant role of the defence component within this sector, and ongoing government support, all corroborate Spain's important presence in aerospace, an area in which it is continuing to grow and develop.



SPAINSAT NG-I Communications Satellite, Toulouse. Image: HISDESAT

THE AERONAUTICAL SECTOR IN SPAIN: GENERAL ASPECTS

Although the 2019 NASS identified aerospace-propagated epidemics and pandemics as a challenge, the emergence of COVID-19 generated an enormous impact on the airspace sector. This pandemic spread rapidly throughout the world and provoked a drastic reduction in air transport, affecting passenger traffic in particular, as authorities attempted to contain the spread of the virus.

Nevertheless, the volume of air traffic recovered surprisingly quickly. But this outcome makes it even more necessary to continue promoting coordinated action among all levels of public administration to protect health and safety and to maintain the safe use of airspace, while also fostering the development of this strategic sector.

During 2024, the aeronautical sector enjoyed year-on-year growth of 8%, surpassing €15 billion in revenue. Looking forward, the period between 2025 and 2030 will be of crucial importance; a strong recovery in commercial aircraft orders is expected, and new models will be developed globally. In this respect, Spain's aerospace industry will seek to effectively position itself, particularly in new technologies focused on emissions reduction.

As with any change, there are significant risks associated with this outlook, but also opportunities in a sector that has traditionally been technologically conservative due to issues associated with security and the costs of certifying new systems and equipment.

In the following, we summarise the challenges our aeronautical industry must face in the coming years:

- Cybersecurity and resilience, within an environment in which cutting-edge technologies coexist with older ones designed for frameworks where cybersecurity was not a default requirement.

- Increase participation in the most profitable commercial aircraft programmes and those of interest for national defence.

- Maintain the capacity to cover the entire aircraft design and manufacturing cycle.

- Maintain leadership in the development and production of air traffic control centres, and of detection and tracking systems (radars).

- Adopt positions to effectively exploit new aviation decarbonisation technologies (particularly hydrogen and SAF), to develop high-energy-density batteries and to make use of hybrid propulsion.

- Foster Spanish involvement in drones and counter-drone systems: this is one of the main growth hubs of the global aeronautical industry.

- Diversify and increase Spanish industrial capacity in systems and equipment beyond aerostructures.

- Maintain Spain's leadership in priority technologies such as composite materials.

- Integrate capabilities across the entire value chain: Primes, Tier 1, SMEs, technology centres, IPOs, universities, etc.

In Spain, the Aeronautical Technology Programme (PTA) has significantly boosted the national industry, providing €190 million in subsidies (2021-2023), and enabling it to participate in the development of key technologies, such as zero-emission aircraft and on-board automation). However, further input is required due to the financial needs and time-consuming nature of R&D investments in the sector.

Neighbouring countries have specific long-term programmes to promote major R&D projects, helping position their companies and organisations for future industrial developments. Therefore, Spain needs a mechanism to continue the PTA and maintain institutional support for the national industry, thus consolidating and enhancing the strategic position gained over the years.

With respect to the production and use of uncrewed aircraft, the entry into force of Royal Decree 1036/2017 and the publication of EU regulations 2019/945 and 2019/947 led to exponential growth (over 1000% in the period 2020-22) in the number of registered operators. By the end of 2024, the State Aviation Safety Agency (AESA) had registered 119,712 uncrewed aircraft system operators in Spain, an increase of 27% compared to 2023. The sector is expected to continue growing following the entry into force of the new Royal Decree 517/2024, which clarifies the scope of application of national regulations and aligns them with European standards.

The above regulatory changes have significantly boosted the sector, improving competitiveness, promoting the creation of highly qualified jobs and reducing its environmental impact.

Figure 6: A400M, EF2000 and F-18 refuelling. Image: Spanish Air and Space Force



The dynamism of the aviation sector and its constant technological evolution are dependent upon effective public-private collaboration, with a significant role being played by the public business sector and with the active involvement of public authorities. Another essential factor is that of coordination between the Ministry of Defence and the Ministry of Transport and Sustainable Mobility, within the scope of their powers, as provided for in Article 6 of Act 21/2003, on Aviation Safety and Security.

In conclusion, the aviation industry is a strategic industrial sector, vital to the national economy, which makes an immense contribution to security and defence, plays a driving role in the innovative ecosystem and offers great potential for transformation. Therefore, it requires the specific attention provided for in Article 10 of Act 36/2015 on National Security.

THE SPACE SECTOR IN SPAIN: GENERAL ASPECTS

Spain is a member of several international organisations related in various ways to space activities. It participates in major European and international space projects and has created important national and international space infrastructure projects.

The Spanish space industry has extensive technological and industrial capabilities in all areas of activity (flight, ground, launchers), ranging from the manufacture of instruments and complete payloads to their integration into complex systems (platforms, satellites, capacity exploitation centres, operations centres, etc.). This capability allows it to participate in the development of complete missions, spanning the entire value chain of the sector. The industry is also present in the applications and services market.

Thanks to the steadily growing output of this industry, which at first was primarily supported by public investment but is now increasingly open to private-sector participation, Spain has its own space capabilities in space surveillance and tracking, telecommunications and Earth observation.

Furthermore, Spain has access to capabilities developed through the components of the European Union Space Programme (EUSPA) and the programmes of the European Space Agency (ESA). Also notable is Spain's participation in the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT), via AEMET (the Spanish Meteorological Agency).

The national capacity for space surveillance and tracking continues to be strengthened, with improvements in the Spanish Space Surveillance and Tracking Operations Centre (S3TOC) as our contribution to the EU space surveillance system (EU Space Surveillance and Tracking), set up to track the trajectories of objects in space, their manoeuvres and potential atmospheric re-entries, to study fragmentation events, to prevent collisions and to provide support for launches into space.

Thanks to the above capabilities, Spain is one of the few nations with the technological, industrial and organisational capacity to form part of multinational surveillance and tracking systems and to help develop essential catalogues of space objects in orbit. These actions reinforce Spain's role as a major player in space security and situational awareness. COVE, Spain's Space Surveillance Operations Centre, part of the Space Operational Command, has additional capabilities for characterising space assets and monitoring irresponsible activities in orbit. All of these contribute to enhance space domain awareness (SDA).

The Spanish space industry is of strategic importance in global terms, due to its specific weight in overall industrial production, the driving role it plays in the innovative ecosystem and other industries, its capacity to transform the economy and the labour market, and its strategic contribution to preserving public safety.

The Spanish space industry is economically vigorous, with revenues of €1.2 billion in 2023, equivalent to 1.2% of the national industrial GDP. The sector is highly productive and export focused, investing strongly in R&D&I and generating a considerable multiplier impact on the Spanish economy.

In March 2022, to ensure the Spanish aeronautical and space industry would continue to play a key role in addressing the challenges and opportunities associated with the major transformations anticipated in the sector in the short and medium term, the Spanish Government approved the Strategic Project for Aerospace Economic Recovery and Transformation (PERTE). This project, which has achieved a high rate of budget execution, was expected to invest €4.533 billion between 2021 and 2025, with initial contributions of €2.193 billion from the public sector and €2.34 billion from private investment.

The Aerospace PERTE and the 2024 Space Technology Programme (PTE) are interconnected as part of a comprehensive strategy to reinforce the space sector in Spain. The PTE is an initiative of the Ministry of Science, Innovation and Universities designed to foster the development and competitiveness of the national space sector. Specifically, it aims to finance R&D projects in the sector to increase the competitiveness of its productive fabric, focusing on the productive capacity of satellite constellations and advancing the maturity of technologies with high disruptive potential, generating activities both in the academic world and in business sectors. As an additional cross-cutting impact, it will support actions targeting the space and the aeronautic sectors.

In 2023, the Statute of the Spanish Space Agency (AEE) was approved, with a component dedicated to National Security. Its mission is to lead space-related projects, to efficiently coordinate the various national departments with responsibilities in the space sector, and to unify areas of international collaboration and coordination within its scope of responsibility. The AEE, attached to the Ministry of Science, Innovation and Universities and the Ministry of Defence, will contribute to organising responsibilities and establishing a national policy to regulate the activities of the public sector in space and to guide the private sector.

The purpose of the AEE is to use space for the benefit, knowledge and security of the Spanish population, and to establish, promote and coordinate all activities and policies that enable research, technological and industrial development and innovation in this field. Its priorities include achieving and maintaining a high level of safety and security

in the operation and functioning of space systems, implementing user segments for optimum exploitation, and developing the applications required for this purpose.

Thus, within the framework of the future National Space Strategy, and with appropriate coordination with other competent bodies, the AEE will support actions aimed at achieving the space-related security and defence objectives set out in the NSS, the EU Space Strategy for Security and Defence (EU SSSD), and similar documents drafted within the NATO framework.

The Air Force's renaming as the Air and Space Force (EA), approved by Royal Decree 524/2022, highlights the importance assigned to matters relating to space, within the framework of security and defence, and underscores the essential role that the Armed Forces must play in the protection and control of space, with the primary involvement of the EA in this environment.

Thus, the Spanish armed forces have adapted their structures and activities to address perceived vulnerabilities in the space domain and to foster a secure environment to protect national interests. More specifically, the EA will ensure free access to and exploitation of space, and protect aerospace infrastructure, resources and services from any challenge or threat.

As part of the above-mentioned adaptation, a Space Command was created within the EA to centralise the training of its units and to implement the direction, planning, organisation and coordination of the EA functions that enable surveillance, control and operation in space.

REGULATORY FRAMEWORK FOR AIRSPACE SAFETY AND SECURITY

Two fundamental laws, the Air Navigation Act 48/1960 and the Air Safety Act 21/2003, continue to shape the legal framework for managing Spanish airspace. Both laws, which implement ministerial powers, have been updated numerous times to adapt to the ongoing evolution of the aviation sector and are entering the final stages of the legislative process for further amendments.

In the European context, the basic references in this respect are Regulation (EU) 2024/2803 of the European Parliament and of the Council on the implementation of the Single European Sky, and Commission Implementing Regulation (EU) 2023/203 laying down detailed rules for the application of Regulation (EU) 2018/1139 of the European Parliament and of the Council with regard to requirements for the management of risks related to the security of information likely to have an impact on aviation safety (PART-IS).

The fundamental task of Spain's law enforcement forces is to protect the free exercise of rights and freedoms and to guarantee public safety and security. This precept is set out both in the Spanish Constitution and in Organic Law 2/1986 on Law Enforcement Forces. The second of these goals, public safety and security, inescapably requires the adequate protection of airspace. In this respect, the responsibilities of law enforcement bodies include public safety at airports, the custody and protection of airport infrastructure and the supervision of compliance with immigration regulations and with the State's fiscal safeguards.

Furthermore, Organic Law 5/2005 on National Defence and the aforementioned Act 21/2003 develop the powers of the Ministry of Defence and the missions of the Armed Forces in the aerospace field, establishing that the Ministry of Defence is responsible for the following tasks:

a. The surveillance, control and defence of Spanish sovereign airspace, and the control of general air traffic in times of armed conflict, as the primary entity responsible for Spain's air defence.

b. The management and control of operational air traffic.

c. The control of general air traffic and the establishment of measures affecting the structure, management, and conditions of use of airspace when so determined by the President of the Government or the Minister of Defence due to extraordinary circumstances or emergency situations that so advise.

The Ministry of Transport and Sustainable Mobility is responsible for the regulation and control of general air traffic in peacetime and in crisis situations other than those involving the Ministry of Defence.

In peacetime, Spain's air defence capabilities are integrated into NATO's Integrated Air and Missile Defence (IAMD) system. As part of our contribution to Alliance operations, Spain offers air defence capabilities outside the national territory.

Among these activities, the Air and Space Force continuously monitors the airspace under its responsibility, employing the human and material resources of its Aerospace Surveillance and Control System (SVICA), Quick Reaction Alert aircraft, the resources provided by the Army and Navy, and the command, control and communications systems that enable the mission to be accomplished. This endeavour, carried out 24 hours a day, every day of the year, employs more than 5,000 personnel and a large part of the EA's infrastructure. Among its duties, air defence protects the national territory against threats such as the renegade use of civilian aircraft for terrorist acts.

Regarding sovereign airspace, its horizontal delimitation is well defined, but not its vertical delimitation, as there is no international agreement on the question. The 1944 Chicago Convention does not contain a definition of airspace, although based on the ICAO definition of aircraft, "*any machine capable of being supported in the atmosphere by the reactions of the air other than those of the air itself upon the Earth's surface*", it could be considered that space begins at the maximum altitude at which aerodynamic flight is possible. None of the five principal treaties on outer space regulate its delimitation.

The European Aviation Safety Agency (EASA) has proposed a roadmap for operations above flight level (FL) 550 (17 km altitude) and does not set a precise upper limit, giving a range between 80 and 120 km altitude. EASA identifies the preparatory actions required in a future regulatory framework for higher airspace operations (HAO).

Twenty-seven categories of future HAO vehicles and operations have been identified. Some are covered by the Chicago Convention and EU Civil Aviation Regulations, others are classified as space operations and fall under the jurisdiction of Member States, and others have hybrid characteristics and require a case-by-case study.

Different types of aircraft and vehicles may coexist within the HAO context, some of them with hypersonic or suborbital capabilities (spacecraft currently under the responsibility of national entities). This means that their residence times within airspace may be very short, and therefore harmonisation or coordination with potential regulations by third parties would be desirable.

In short, while the EU treaties and basic acts allow some of these operations and assign the EU shared competence to regulate some of them, particularly those classified as civil aviation operations carried out within the EU, most of the current implementing rules would need to be adapted or new rules would be required, for example in areas concerning airworthiness, operations, navigation and air traffic management, environmental issues, airfields, personnel licensing, etc.

Furthermore, given that some of these operations will be uncrewed, synergies with drone regulations will also need to be further evaluated. The safety and security risks of these operations while transiting current airspace below FL 550 and when crossing into the higher airspace will need to be taken into consideration.

In the civil sector, matters regarding uncrewed aircraft are regulated by EU regulations (Delegated Regulation (EU) 2019/945 and Implementing Regulation (EU) 2019/947). These are complemented at the national level by Royal Decree 517/2024, which supplements European regulations in their areas of application and updates national regulations in the areas excluded from the EU standard (military activities,

customs, police, search and rescue, firefighting, border control, coastal surveillance, and similar). In addition, Royal Decree 517/2024 develops the U-Space regulatory framework (a set of specific services and procedures designed to ensure safe, efficient and affordable access to airspace for drones), which is also regulated at the European level by Implementing Regulation (EU) 2021/664.

Royal Decree 517/2024, which develops the legal framework for the civil use of uncrewed aircraft systems (UAS), specifies that, for reasons of public safety, all drone operations (excluding those carried out in the execution of non-EASA regulated activities or services) carried out over urban environments or in crowded areas must be reported to the Ministry of Home Affairs. In this regard, drone operations may be limited or prohibited when they may give rise to serious risks to the protection of persons or property.

Similarly, in order to prevent, investigate or detect the commission of criminal and/or administrative offences, including the protection and prevention of threats to public safety, the Ministry of Home Affairs has created a Registry of Uncrewed Aircraft and of their Marketing and Sales, applicable to all uncrewed aircraft with a maximum take-off mass (MTOM) of 250 g or more or that are equipped with a sensor capable of capturing personal data, regardless of their MTOM, unless it complies with Royal Decree 1205/2011, on the safety of toys, and which compiles data that ensure the traceability and identification of aircraft and their owners. This measure strengthens the necessary control of uncrewed aircraft, whose negligent use can have serious consequences for public safety.

Under the powers conferred by Organic Law 2/1986, on Law Enforcement Agencies and Bodies, and to ensure the free exercise of rights and freedoms and the protection of public safety and security, the Ministry of Home Affairs makes use of the SIGLO-CD application, a system specialised in the detection, control and neutralisation of drones. This system fulfills both the functions of the above law enforcement agencies to exercise administrative control over the professional, recreational and sporting use of drones, and also constitutes a fundamental instrument in the fight against the use of these aerial platforms by organised crime and terrorists.



Figure 7: Members of the National Police and Civil Guard with drone jamming systems. Image: Ministry of Home Affairs

REGULATORY FRAMEWORK FOR SAFETY AND SECURITY IN SPACE

The lack of adequate legislation represents a significant challenge to the governance of space and to the protection of national interests. Currently, there is intense competition for licences to use space frequencies in orbits, especially low earth orbits (LEO), which provide nations with strategic and operational superiority. The growing participation of non-governmental entities in the sector (known as New Space) highlights the urgency of establishing a national regulatory framework that regulates space activities, ensures the necessary legal certainty, and aligns the use of space with national priorities, while facilitating legislative development at the international level.

The main space treaties date back to the Cold War, with the 1967 Outer Space Treaty being the cornerstone of space governance. Spain has ratified key agreements, such as the 1963 Moscow Treaty, the 1968 Agreement on the Rescue of Astronauts, the 1972 Convention on International Liability, and the 1974 Convention on the Registration of Space Objects. A new Royal Decree is currently being processed to update Royal Decree 278/1995, which establishes the Registry provided for in the 1974 Convention on the Registration of Space Objects in Spain and adapts it to the evolution of the sector.

At the international level, the United Nations Office for Outer Space Affairs (UNOOSA) and the Committee on the Peaceful Uses of Outer Space (COPUOS) coordinate peaceful cooperation in the use of space. Furthermore, the Space 2030 Agenda promotes strategic partnerships for sustainable space development.

To mitigate the problem of space debris, the Inter-Agency Space Debris Coordination Committee (IADC) recommends measures such as the "25-Year Rule" and sustainable satellite design.

Spain has been a signatory to the Artemis Accords since 2023, which establish principles for responsible space exploration, including transparency, the peaceful use of space and the safe disposal of space vehicles. Adherence to these principles makes operations in the space environment safer and more predictable.

Regulation (EU) 2021/696 set up the EU space programme for 2021–2027 and the EU Agency for the Space Programme (EUSPA), in order to enhance the competitiveness of European countries in the sector and harness the potential offered by space for the security of the Union and its Member States. The Regulation establishes the principles of governance and action for each of the European organisations involved, defining their functions and promoting a high degree of coordination.

The EU Space Strategy for Security and Defence underscores the importance of developing international legal instruments for the global governance of the use of space. The Strategy also emphasises the need to periodically update space-related threats, via the near-real-time collection and analysis of security incidents that may be indicative of such a threat. It also highlights the need to enhance the Union's capacity to respond to space threats by improving information exchange and threat response procedures, conducting exercises to validate these procedures, and creating synergies among the instruments available.

The European Commission, under the leadership of the newly-appointed Commissioner for Defence and Space, has developed a proposal for a Regulation on the safety, resilience and sustainability of space activities (the "EU Space Act") to provide a regulatory framework to support safer and more efficient management of space systems, protect space infrastructure from threats, incorporate mandatory safety and cybersecurity requirements for space assets, introduce common standards, harmonise requirements for space operations licences, promote the sustainability of space operations, and boost economic growth and services in space.

In line with the above, Spain's future legislation on space activities, currently being drafted by the Spanish Space Agency, will determine, among other questions, the security requirements and activities necessary to regulate space activity. This law will enable us to defend our interests within a context of cooperation and to meet our international obligations, whilst offering security to all stakeholders, both public and private.



Figure 8: SPAINSAT NG-I Communications Satellite, Florida launch. Image: HISDESAT

THE NATIONAL AEROSPACE SECURITY STRATEGY, STRENGTHENING OUR SECURITY

TWO DOMAINS, A SINGLE SECURITY STRATEGY

Any object that could reach the Earth's surface from space and pose a threat would inevitably have to transit through the air from the orbital space of origin. Therefore, the functions of surveillance, detection, identification and classification of such objects must be extended to determine the most appropriate response, as well as to collaborate in the attribution of actions that, intentional or otherwise, endanger aerospace security.

Advances and falling costs in space technology have led to a proliferation of private companies and public-private partnerships with different interests in space. These partnerships have exponentially increased the number of annual operations carried out in this field, while simultaneously increasing the threats posed to both aviation and space operations.

With ever more objects orbiting or transitioning from orbital regimes to controlled airspace, a higher degree of coordination will be required to minimise the risk of collision, and more manoeuvres to avoid impact will be needed, whenever this is an option. In both cases, the consequences are negative, as they can lead to increased restrictions on the use of airspace, generate difficulties in coordination mechanisms and reduce the useful life of orbiting systems.

In the other area of interest, threats to space-based positioning, navigation, timing and communications systems (which are essential to the operational safety of air traffic management) make it essential to conduct a comprehensive view of aerospace security, while recognising the specific characteristics of the air and space domains.

To address threats to the integrity, functionality and sustainability of the air and space domains, the National Aerospace Security Strategy includes measures, policies and coordinated actions designed to mitigate such threats, ensure freedom of action in air and space, and protect the systems and capabilities operating in these domains.

The availability of a single National Aerospace Security Strategy, establishing security objectives and measures for both domains, has a catalytic effect, increasing the effectiveness of the measures implemented in monitoring, controlling and responding to incidents.

STRATEGIC APPROACH TO AEROSPACE SECURITY IN SPAIN

The main national interests subject to protection in the air and space domains are:

- The safety and well-being of the Spanish population, on Earth, in the air and in space.
- Airspace subject to Spanish sovereignty, responsibility or national interest, regarding unauthorised incursions of any kind.
- A solid technological and industrial base of high strategic value.
- Critical aerospace infrastructure, particularly aerospace surveillance, control and defence systems.
- Essential products, applications and services provided by the aerospace sector.
- Freedom of air navigation and the safety of aircraft transiting through airspace under national responsibility.
- Access to airspace for Spanish aircraft and for airspace-related security and defence facilities.
- Free access to space, and freedom of action in the operation and exploitation of space to ensure the continuity of space capabilities and services of strategic national interest.

- Radio frequencies allocated by the International Telecommunication Union to national space assets, associated orbital positions in the geostationary satellite orbit, and associated characteristics of satellites in other orbits.

- Compliance with international treaties regarding aerospace safety and, in this regard, commitment to the international organisations of which Spain is a member.

- The development of launchers, space and ground infrastructure, and aerospace data and services centres.

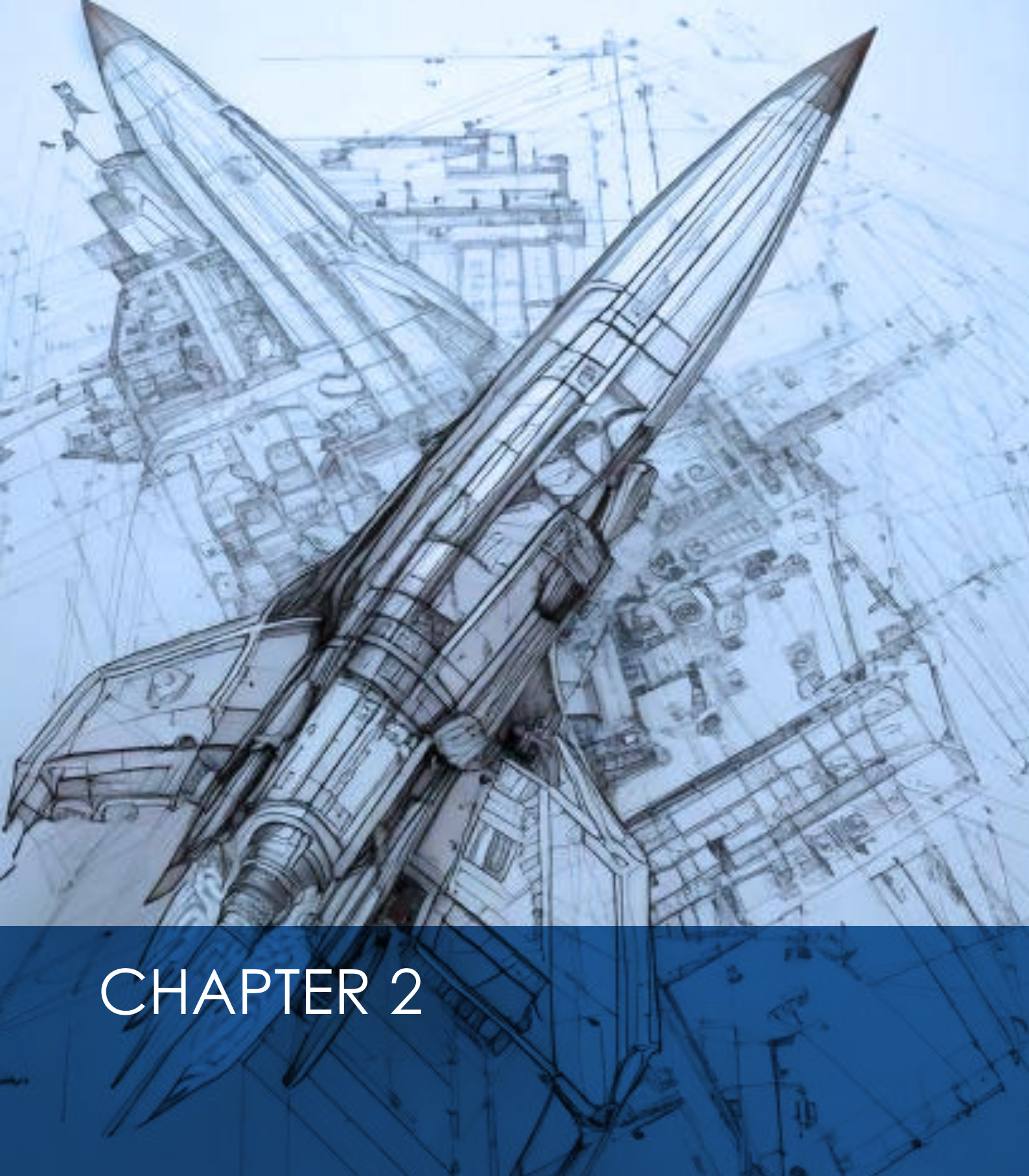
- Protecting public health against the spread of pathogens or toxic substances, intentional or otherwise, by aerospace means.

- The promotion of scientific and technological knowledge in the aerospace field, fostering the development of our own technologies and reducing dependence on other countries for the supply of critical components and essential aeronautical capabilities to ensure the greatest possible sovereignty and strategic autonomy.

- The environmental sustainability of aerospace operations.



Figure 9: HISDESAT Satellite Control Antenna. Image: HISDESAT



CHAPTER 2

THREATS TO NATIONAL SECURITY IN THE DOMAINS OF AIR AND SPACE

INTRODUCTION TO THE CONCEPTS OF RISK AND THREAT IN THE DOMAINS OF AIR AND SPACE

The 2019 National Aerospace Security Strategy defined the concepts of threat and challenge, distinguishing the two according to intent. However, in the present review of the question, our approach to these definitions avoids any initial attempt to assess the intent or otherwise of an event due to the difficulty of unequivocally demonstrating this in certain scenarios.

For the purposes of this Strategy, therefore, "Threat" is defined as a situation or event, whether natural or of human origin, and whether accidental or deliberate, that, if materialised, could cause damage or loss to the assets to be protected.

Accordingly, in the domains of air and space, threats are considered transversal to the contexts of safety and security.

By contrast, "Risk" is defined as the result of combining the expected probability and impact on an asset to be protected, should a threat materialise. This approach facilitates risk analysis and management, and subsequent decision making, enabling an appropriate, prioritised and anticipated response to be made. In risk management, the risk level is determined from the combination of probability and resultant impact.

In view of the current proliferation of dual-use technologies, specific mechanisms are needed for determining threats, considering not only the presence of the technology, but also the behaviour of the parties involved. Services that are essential to public safety and well-being and to the security of air operations may depend on space assets, which in certain cases adds complexity to categorising threats as exclusive to the air or space domain. Moreover, the assets operating in these areas may present vulnerabilities.

Although the Strategy addresses the specificities of the air and space domains separately, in terms of threats they must really be considered interconnected domains, due to their nature and because of their shared vulnerabilities to transversal effects in certain cases. In the following, we present a non-exhaustive list of threats that, while each is unique and has a different nature and origin, are common to the domains of air and space.

ARMED CONFLICTS

Growing strategic rivalry and economic decoupling are increasing global instability and heightening competition for control of resources, which could lead to an escalation of the use of force and give rise to armed conflict. The lack of effective peaceful resolution mechanisms, together with the rise of unilateralism, generates an environment of mistrust, which increases the risk of a war between major powers due to miscalculations or misunderstandings.

Armed conflicts, in addition to the obvious threat they pose to the aerospace capabilities involved, also endanger other assets that share the same physical operating environment and may suffer the collateral effects of operations, even if they take place in remote areas. Such actions can result in the closure of airspace or the disruption of international routes, requiring the redesign and coordination of flight routes to avoid risks. Space, too, has become a vulnerable strategic domain, where satellites are exposed to interference or attack, highlighting the need to strengthen international cooperation in the protection of these assets.

Figure 10: EF-2000 Typhoon (C.16). Image: Spanish Air and Space Force



The following are some potential threats to aerospace systems in the context of armed conflict:

Kinetic physical attacks against air/space systems (for example, anti-satellite weapons) or against the infrastructure from which they operate (bases and airports, command and control centres, ground-based control and tracking centres, spaceports). These attacks seek the partial or total destruction of the target through direct impact, temporarily or permanently affecting its capacity. They may be carried out using not only weapons designed for this purpose, but also weapons capable of accurately reaching objects in space from the ground.

Non-kinetic physical attacks against air/space systems or the infrastructure that enables their operation and control. These attacks, while not seeking direct physical contact with the target, aim to cause physical damage, degrading its capabilities. They may be carried out using directed energy weapons (lasers, particle beams, radio frequency), and used from the ground, in the air or from space. Depending on the characteristics, they can cause temporary or permanent damage to the systems targeted.

Electromagnetic attacks that use energy to secure the use of the electromagnetic spectrum for their own benefit and deny it to the adversary. These attacks, as in the previous cases, can affect air and space systems and ground-based infrastructure, including the radio frequency links that facilitate their capabilities.

This category includes various techniques such as spoofing (capturing, altering and retransmitting a signal to confuse the receiver) and interference or jamming (masking or obstructing the signal at the receiver). These types of techniques are frequently used to disrupt communications and PNT (positioning, navigation and timing) services. The degradation or unavailability of these services may impact on all areas of national security.

While the nature of these attacks does not directly seek the physical destruction of the target, this outcome often occurs, potentially causing loss of human life and the loss of the asset itself. Moreover, after the asset is captured it may be subjected to reverse engineering.

Nuclear attacks; such attacks generate an electromagnetic pulse that affects all electrical components, and have great physical destructive power with devastating short-, medium-, and long-term consequences for the affected area and its inhabitants. If one of these weapons were detonated in space, the impacts would be indiscriminate on multiple satellites, potentially generating a chain reaction (the Kessler effect) with unforeseen consequences for the interests of the launching State, and possibly catastrophic repercussions for the entire Earth.

Weapons of mass destruction can use aerial means to transport an attack or to serve as a vector for this purpose. In this respect, drones, at relatively low cost, can reach targets that were previously limited to missiles.

While the materialisation of one or more of these threats is a real possibility in the context of armed conflicts, they may also occur in other scenarios, given the existence of non-State actors with solid sources of financing and access to capabilities (including nuclear or other agents of mass destruction) that can be used against certain aerospace assets and the facilities that support their operation.

In response to these dangers, the Spanish Air and Space Force provides ongoing protection for national airspace through its air defence systems, based on continuous surveillance (24/7) through a network of radars, command and control centres, and armed combat aircraft on permanent alert, ready to identify, intercept and neutralise any aerial threat that may arise. This air defence system forms part of NATO's Integrated Air and Missile Defence System, via the Combined Air Operations Centre, located at Torrejón de Ardoz Air Base (Madrid), which controls this mission for all of southern Europe, from the Canary Islands to Türkiye.

THREATS FROM HEALTH-RELATED EVENTS

The World Health Organisation defines an epidemic as a disease that spreads rapidly and actively, causing a significant increase in the number of cases, but remains confined to a specific geographic area. Yellow fever, smallpox, measles and polio are excellent examples of epidemics. A pandemic is the global spread of a disease that in addition to the above, meets the following two criteria: it must affect more than one continent and the cases in each country are not necessarily imported but can be caused by community transmission.

Throughout history, a considerable number of pandemics have been recorded. These events have not only impacted on public health but have affected politics, the economy and society at large. Moreover, major health threats have also prompted the development of landmark public health measures, such as the establishment of quarantines and the development of effective drugs and vaccines.

The COVID-19 pandemic had a significant impact on the aviation industry due to the travel restrictions implemented worldwide, resulting in fleets of parked aircraft, empty airports and a sharp drop in passenger numbers (over 90%). The sector, thus, faced a completely new scenario. In Spain, as elsewhere, a multisectoral and multidisciplinary approach was taken to disease intervention and control to mitigate



Figure 11: Health Checks at Adolfo Suárez Airport. Image: Ministry of Health

the effects of the virus in all sectors of society, including the aeronautical industry. The implementation of health controls at airports enabled cases to be identified at these border entry points, as part of the epidemiological surveillance systems established. The effective collaboration between the aviation and healthcare sectors facilitated the gradual resumption of international travel to Spain, in line with the reduced intensity of the pandemic.

Current early warning and rapid response systems enable the prompt identification of serious cross-border threats to public health, using epidemic intelligence as a tool for detecting potential events of concern. Once assessed, these events are categorised according to their risk level, in order to prioritise the types of intervention applied and to implement specific actions of prevention, mitigation and control. These systems, which are of vital assistance to the healthcare sector, enable threats to be monitored in real time and inform the appropriate actions to be taken in each case. The world's experience with the COVID-19 pandemic has shown that events of such magnitude have significant repercussions for society as a whole and therefore require joint intervention for their prevention and control.

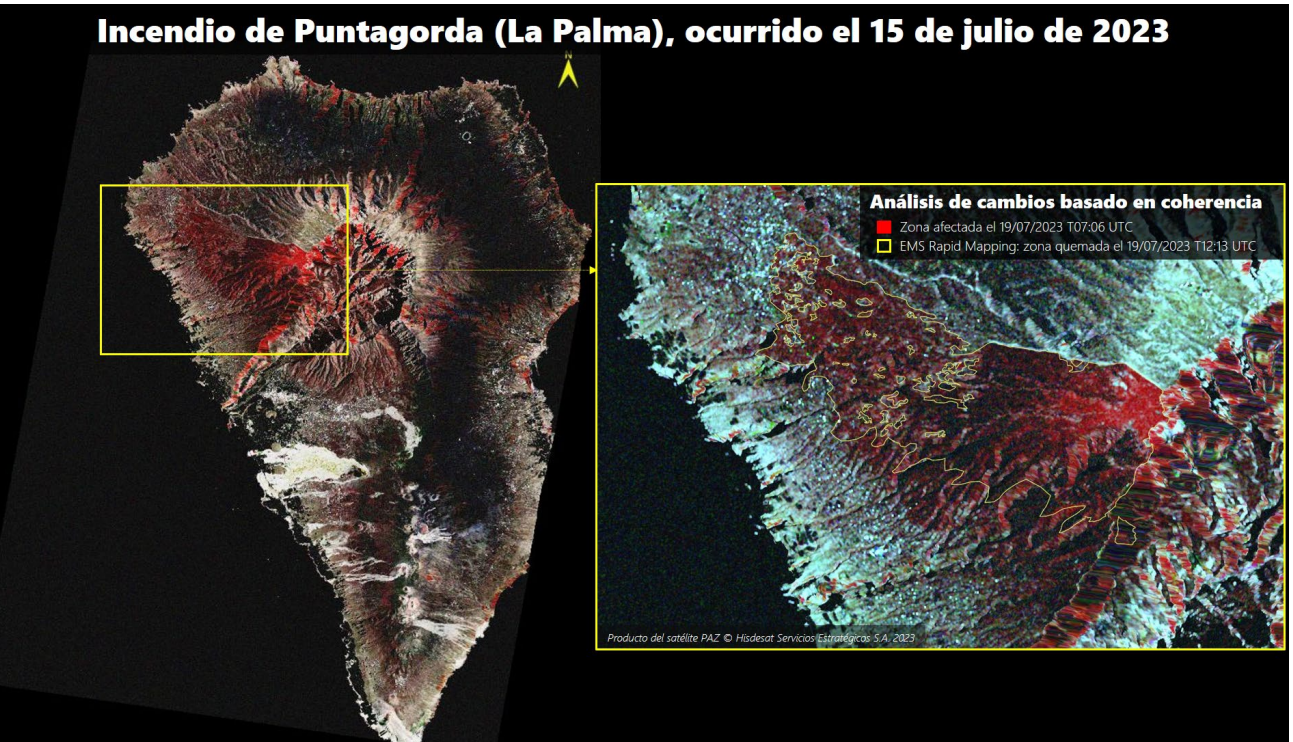
NATURAL CATASTROPHES

Natural disasters pose a significant threat to the aerospace sector due to their ability to disrupt operations, damage critical facilities and operating personnel, and jeopardise the safety of aircraft and space systems.

Infrastructure located on the Earth's surface is vulnerable to various natural phenomena, both geophysical and meteorological in origin. The former include earthquakes and tsunamis, while the latter include floods, intense rainfall caused by high-level depressions (known as DANAs in Spain), hailstorms or very strong winds, and tornadoes. The damage caused by these phenomena can paralyse aerospace operations for days or weeks, causing significant economic losses and disruptions to the global supply chain (fuel, electricity, materials, etc.).

Other types of natural phenomena, such as volcanic ash, thunderstorms, frost and fog, can directly affect the safety of aerospace operations.

Figure 12: Satellite image of the PAZ fires in La Palma, 2023. Image: HISDESAT



In addition to those mentioned above, threats may be produced, directly or otherwise, by adverse space weather phenomena, originating outside the atmosphere. Another type of threat is that posed by the approach of celestial bodies to Earth (Near Earth Objects) such as asteroids or other objects, which can collide with space systems in orbit or re-enter the atmosphere with the consequent danger to the safety of air operations and even to the area of the Earth's surface where they ultimately impact.

Adverse space weather phenomena include solar flares, coronal mass ejections and high-speed solar winds. These events, with concurrent geomagnetic storms and atmospheric disturbances, can cause interference in communications, navigation and power distribution systems.



Figure 13: Recreation of a solar storm

Space Weather Advisory Information Centres, whose tasks are defined in ICAO Annex 3, provide worldwide updates on space weather conditions for commercial and general aviation. These centres issue the corresponding advisories on space weather that may affect high-frequency communications related to air transport, navigation and surveillance based on global navigation satellite systems (GNSS), and radiation levels on board civil aircraft.

SUPPLY CHAIN INSECURITY

Some nations have developed strategies aimed at blocking or restricting third-party access to certain technologies or raw materials, often essential for supply chain procurement. This has led to abuse of their monopolistic position over specific services and products (e.g., rare earths used in high-tech equipment), or to their preventing the export of components.

This scenario penalises the competitiveness of the national and European aerospace industry due to its inability to find substitutes for certain materials or to the denial of permits to re-export certain technologies, with consequences that may be particularly severe in the current complex geopolitical situation.

Boosting domestic industry and imposing export controls might be key policies to successfully negotiate and overcome permit denials. Furthermore, classifying certain aerospace technologies as critical to national security would strengthen negotiations in the face of supply blockades, improving production and reducing dependence on other countries.

It is also essential to acquire the greatest possible strategic autonomy regarding the supply of critical components and vital aeronautical capabilities. Accordingly, Spain should produce a catalogue of technologies of strategic importance for its internal development, and launch innovation programmes to achieve sovereignty and autonomy in these areas, in conjunction with associated international programmes at the European level.

TERRORISM, VIOLENT EXTREMISM, ORGANISED CRIME AND MAJOR FELONIES

The threat of terrorist attacks against the commercial aviation sector, military aerospace capabilities and/or their infrastructure, while always present and never out of consideration, is especially acute in the current context, in which there has been a significant increase in the overall terrorist threat, mainly arising from the conflict in the Middle East. Global terrorist organisations are now particularly active, attempting to plan and execute high-impact actions.

Among such dangers, plots against the airline industry are once again at the forefront of terrorists' ambitions. The September 11 attacks in the United States in 2001 clearly demonstrated that attacking the airline industry is one of the most profitable targets for terrorist groups, both in terms of operational success, in the number of casualties produced, and, above all, due to its capacity to generate terror and the need for multimillion-dollar investments to improve the sector's security.

Jihadist groups have had, still have and will continue to have a primary interest in planning and executing attacks against the commercial aviation sector, and their techniques, tactics and procedures have been modified over time and adapted to the circumstances, capabilities and vulnerabilities detected at each time and place.

Airspace can be exploited by terrorist groups to carry out attacks, whether through the hijacking of aircraft or through the use of aircraft for terrorist acts. However, the Spanish Security Forces are equipped to identify potential terrorist threats involving the use of airspace and neutralise them before they occur. Collaboration with intelligence agencies and international cooperation are key factors in preventing such attacks. The development of specific intelligence on air threats, the effective use of early warning systems and the performance of joint exercises by the National Police, the Civil Guard and the Armed Forces are necessary to ensure a rapid and coordinated response.

Some violent extremist groups seek to disrupt good order and security in airport environments by breaching perimeter fences and entering restricted security areas, possibly gaining access to the apron where aircraft are parked, in order to carry out protests. For the purposes of this Strategy, these types of actions could be considered

violent extremism; while not related to terrorism, they impact on airport security and, consequently, on civil aviation.

The use of airspace for illegal activities, such as drug trafficking or smuggling using light aircraft or drones, is a constant risk that has been observed and confirmed in the activities of organised crime.

The use of drones has expanded dramatically in recent years, in number and in diversity. The mere presence of drones in the vicinity of aerospace, civil and military infrastructure poses a threat to operations, regardless of the user's purpose. Currently, terrorist groups use drones for reconnaissance and propaganda purposes, as transports for explosive devices to be released in the direction of the target, and as attack vectors (suicide drones). To date, the drones used by terrorists have mostly been recorded in conflict zones, but their use elsewhere in the West is considered only a matter of time.

Commercial drones have been used to smuggle items, primarily drugs and cell phones, into prisons. This facilitates the continuation of criminal activity within prisons and increases the risk of inmate escape. For some criminal organisations, smuggling weapons into prisons using these types of drones, equipped with GPS and other satellite navigation technologies, is a practicable option, and one that would severely compromise prison security. The use of aerospace-based means for escape also poses a significant threat.

To address such threats and strengthen security in air transport and airport infrastructure, the Civil Guard has developed several capabilities in addition to the units it permanently deploys. ÍCARO (Identification of Anomalous Conduct and Operational Response) teams operate at airports to identify passengers who may pose a threat, analysing potentially criminal behaviour and detecting attempted deception. In the air, National Flight Security Escort Units (UNESEV) operate on board Spanish aircraft, both on domestic and on international flights. These specially trained Civil Guard officers, who undergo rigorous training, have the primary mission of preventing and responding to any attempt at unlawful interference with commercial air transport, such as the seizure of an aircraft in order to hijack its passengers or even crash it into a target. Finally, Pegaso teams are deployed throughout the country. Their main function is to prevent the use of light aircraft for illegal activities, such as drug trafficking or smuggling. The Pegaso teams also protect airport environments from the unauthorised use of drones.



Figure 14: Members of the National Police and Civil Guard airport security. Image: Ministry of Home Affairs

National Police stations at airports provide intense training for Central Border Unit personnel serving at security checkpoints and border controls, making them highly skilled at detecting, observing and analysing abnormal passenger behaviour. This training is ongoing and complements the experience, skills and specific knowledge of these personnel, who, over time, have perfected their ability to recognise suspicious behaviour patterns and interpret nonverbal signals. The rapid, accurate detection of these signals enables police officers to identify potential threats and, if necessary, implement preventive measures prior to boarding to avoid incidents on the aircraft. The presence of National Police personnel on commercial flights, in addition to ensuring compliance with immigration laws, contributes to strengthening air safety. Since these officers are specifically trained to operate on aircraft, they play a fundamental role in maintaining order in sensitive situations and providing appropriate response to any attempt at unlawful interference that might jeopardise the flight. In addition, when the Ministry of Home Affairs detects a heightened threat to air safety, such as an attempt to hijack aircraft, or in specific cases that call for intervention, the National Police may deploy specially trained and equipped officers to neutralise criminal activity.

In addition to the above functions, the National Police provide coverage in the area of general civil aviation control through the Air Security and Protection Teams (ASPA) of the Police Air Unit to ensure compliance with aeronautical legislation and to prevent

and, where appropriate, prosecute all types of criminal activities committed by aerial means. The ASPA teams' primary mission is to provide security and protection of airport environments to prevent unauthorised drone flights from affecting the safety and normal operations of air transport.

Spanish law enforcement forces coordinate efforts to monitor unauthorised flights, especially in border areas and near illicit trafficking routes. However, their efforts are hampered by the fact that criminal organisations may use radar-type technologies to locate the presence of law enforcement drones or aircraft. This first-hand information on police deployments enables criminal organisations to evade surveillance, facilitating activities such as drug trafficking, human trafficking and irregular immigration.

New methods of aviation-related illegal entry into national territory have been detected, more sophisticated than those commonly found by maritime routes. For example, taking advantage of the fact that a plane, occupied by passengers from areas outside the Schengen Agreement, is flying over national territory, some occupants might fake a medical emergency, forcing the plane to land at the nearest airport. This exceptional circumstance is then used by passengers to hastily abandon the aircraft once on the ground and avoid police action. This type of behaviour not only compromises the safety of the flight and its passengers, but also that of the airport itself, as it puts both the integrity of the aircraft and the security of the airport environment at risk, Furthermore, these passengers may include potential terrorists or foreign fighters, which represents an additional risk to national security, as they could exploit this vulnerability to enter the country for purposes related to terrorism or subversive activities. While these types of incidents, from a migration perspective, have a low impact, they can seriously threaten infrastructure, even blocking runways. Moreover, coordinated actions, within the framework of a hybrid strategy, could have a significant impact on national security. The possibility of evading rigorous controls incentivises persons seeking to enter the country irregularly, creating a 'pull effect'.

Related to the above, it is also necessary to consider the potential use of private flights to commit criminal offences related to the smuggling of defence material and dual-use goods and technologies. These types of flights are subject to fewer controls than commercial flights; in most cases, only security controls are carried out, not

fiscal controls. Consequently, there is a real risk that they could be used for the above purposes or even as vectors for weapons of mass destruction, if terrorists or other non-State actors were able to transport radioactive or explosive materials as part of an attack.

Although the threat of terrorist and criminal groups directing attacks against satellites is relatively low, they might address critical terrestrial infrastructure in order to more easily and cheaply impact the services provided by space systems.

Use of explosive and incendiary devices in air cargo: This is a new pattern in the hybrid strategies developed by hostile actors and a growing risk to the aerospace security of Spain and its neighbours. Since 2024, and due to the support provided to Ukraine by various countries, including Spain, during the Russian invasion of its territory, the use of explosive and incendiary devices concealed in packages subsequently transported by air, among other means, has significantly increased, representing a serious threat to Spanish aerospace security. This phenomenon is part of the hybrid strategy of State and non-State actors, with a high probability of exploitation by hostile intelligence services.

Among the cases detected in the European Union, some shipments have been found to contain flammable materials and incendiary devices, to be detonated just before being loaded onto the aircraft. Although these hostile actions affect the sector collaterally, the potential risk is very high; if these substances were detonated inside an aircraft in flight, the consequences could be fatal. Unlike traditional attacks against airport infrastructure or aircraft in flight, the use of explosives and incendiaries in air cargo allows attackers to exploit vulnerabilities in the logistics chain to maximise their impact with minimal operational exposure.

Chemical, Biological and Radiological (CBR) threats: The use of chemical weapons in past terrorist attacks against non-aviation targets and on the front lines during recent conflicts has highlighted the potential consequences of chemical agent attacks against civil aviation. Terrorist groups have previously demonstrated an interest in producing and using CBR agents in attacks against the public areas of airports.

CYBER THREATS

The infrastructure of a complex airport environment, including aspects related to terrestrial functions and the centres that manage and regulate air traffic, are a high-value, high-impact target for those who, through cyberattacks, seek to cause chaos or exploit the information stored on their servers. Hence, the work of the agencies responsible for preventing these attacks, as well as quickly restoring damage should it occur, is vitally important.

So far, relatively few attacks have directly targeted civil aviation, but in recent years, terrorists have shown increased interest in cyberattacks and a general intent to carry them out. At present, hacking activities by extremists are characterised by relatively basic denial-of-service attacks. No examples of specific terrorist cyberthreats against aircraft, airports or airbases have been detected, nor are there any indications that terrorist groups have acquired or are developing advanced capabilities or that they currently consider this an effective way to carry out a mass-effect attack, especially when compared to more obvious and proven attack methodologies. Nevertheless, the possibility of such an attack in the future cannot be ruled out.

Air defence command, control and communications systems must address a similar threat, with the added complexity that it may originate from State or non-State actors.

In the space domain, cyberattacks represent a growing threat, capable of seriously affecting both terrestrial infrastructure and orbiting space systems.

Given the hyperconnectivity of our systems and our dependence on networks and information systems, if appropriate protective measures are not implemented the vulnerability to cyberattacks may be high. These could have devastating consequences, compromising the safety, operability and reliability of critical space services and even affect national security.

A cyberattack targeting space assets could disrupt satellite operations, preventing them from fulfilling their function or impeding their control and tracking. This, in turn, could provoke a trajectory alteration, leading to mission failure and the loss of the asset. Moreover, an altered trajectory might pose a risk to other satellites, requiring

operations to avoid collisions; should these occur, the State responsible for the hijacked object might be legally liable for the consequences.

Cyberattacks could also be conducted to steal, leak or manipulate sensitive data from specific targets, endangering national security and compromising strategic information. The alteration of critical data can lead to poor decisions, impacting the safety and efficiency of space operations. The ready access to constantly updated malware vehicles and the integration of generative AI into attacks also broaden the threat spectrum, making it accessible to less skilled actors.

ESPIONAGE

Aerospace espionage is one of the most insidious and difficult-to-quantify, and sometimes attribute, threats, and its effects can be profound and far-reaching. It is not limited to military or government contexts, as a means of obtaining classified or sensitive information, but can extend into business, scientific and research fields through new techniques, tactics and procedures, which in many cases are linked to information technologies.

Espionage can enable the perpetrator to neutralise technological and operational advantages in the national aerospace sector and thus provoke relative disadvantages, especially with respect to disruptive, decisive technologies in fields essential to security and defence.

Multispectral observation satellites can perform the continuous monitoring of ground activities and collect critical data on movements and positions, which is essential for strategic decision-making. In recent years, small satellites in low-Earth orbits have become a valuable resource for the characterisation, monitoring, geolocation and tracking of critical entities.

In response to the current proliferation of satellites and the need to determine their threat potential, specific systems are being developed to observe other objects from space, providing non-earth imaging. In addition, existing capabilities of Earth

observation satellites are being optimised such that when they are not flying over areas of interest, they can direct their sensors toward other space objects to gather information.

Communications interception is another common form of espionage. To combat this danger, communications links with space vehicles must be protected using appropriate encryption elements and, to the extent possible, sensitive information should not be transmitted through infrastructure lacking such protection.

Regarding industrial espionage, measures must be adopted to prevent unauthorised access to knowledge of strategic space developments and classified materials.

In this regard, one of the most common vulnerabilities the sector faces arises from the high level of cooperative development with third countries that may be more industrially advanced and a lack of protection against technological absorption measures developed by these countries. To combat this danger, we must determine which technologies are shareable and which should be considered essential for national security and protected accordingly.

THREATS FROM INCREASED SPACE ACTIVITY

With the introduction of new uses of space, there has been a corresponding increase in the number of actors involved and in levels of space activity, a pattern of change that is expected to continue in the coming decades. The exploration of new ways to monetise space operations is resulting in an entire ecosystem of opportunities together with traditional space-use activities. These new possibilities include space tourism, mining, colonisation and more.

Additionally, within the traditional uses of space, activity is also increasing with the deployment of mega-constellations of mini-satellites, and the resulting increase in the number and types of launches, some of which enable components to return to Earth's surface for reuse.

This increase in space activity poses a threat to the long-term sustainability of space operations if the existing regulatory framework is not adapted to address changing circumstances.

For example, the orbits closest to Earth are now saturated with operational space objects, and also with space debris.

This space 'junk', regardless of its nature, significantly endangers current and future space operations, with repercussions on exploration, research and many other activities. The re-entry of space objects can also affect the safety of operations in airspace and on Earth's surface.

The contamination of space with this debris and the overcrowding of space assets in the most profitable orbits limits compromises the notion of free and safe access to space for all. Furthermore, impacts of space debris on operational assets are occurring, heightening the risk of the Kessler effect (a cascading space debris scenario where the density of objects in low Earth orbit becomes so high that collisions become increasingly likely, leading to a chain reaction of further collisions and debris creation).

The number of objects in orbit already poses a challenge for Earth-based space observation missions, as their reflections and obstacles prevent dark sky observations.

In response, Spain and Chile have created the Group of Friends of the Dark and Quiet Sky, which has been joined by a growing number of countries, to promote dialogue at all levels on the most appropriate measures for preserving dark skies, in the conviction that this initiative has instrumental value for the development of science and is closely linked to the long- and medium-term sustainability of space activities.

Contributing to operational sustainability is a global responsibility, and the measures to achieve this goal offer new opportunities, but also new threats, stemming from the dual nature of many space capabilities.

Advances in in-orbit servicing are revolutionising space asset management, combining technological opportunities with strategic challenges to safety. These operations, designed to extend satellite lifespans or enhance their capabilities, involve technologies such as robotic arms, capture systems and rendezvous and proximity operations (RPOs). However, their dual nature poses geopolitical risks, as the same techniques could be used to alter the trajectories of third-party satellites.

Space intelligence activities have also been developed to observe space assets from space, using space surveillance and tracking systems. These make it possible to track and identify space vehicles for extended periods of time, using orbiting space assets, probably with the intention of carrying out intelligence operations, such as signal capture.

Rising levels of space activity also require enhanced coordination with airspace activity, as additional risks to operations may be can generated if adequate resources are not available to accurately track and monitor space assets as they pass through the atmosphere during exit and re-entry.

The Spanish Space Surveillance and Tracking Operations Centre (S3TOC) and the Space Surveillance Operations Centre (COVE) under Spanish Space Command (MESPA), the S3TSR radar and the associated sensor network are all essential for the surveillance and tracking of objects in space and operate in a coordinated manner, although each has its own specific tasks.

ANALYSING SPACE THREATS AND FORMULATING A JOINT RESPONSE

Spain, through its Armed Forces, the Spanish Space Agency and other administrations, is alert to the vital importance of space and satellite services. The 2021 National Security Strategy identified space as an area of competition, requiring protection. Royal Decree 524/2022 incorporated the term "Space" into the denomination of the Spanish Air Force, highlighting the need to reduce its vulnerability and to develop appropriate surveillance and reconnaissance capabilities. In this respect, Order DEF/264/2023 created the Space Command to address growing international competition and the development of technologies aimed at restricting the use of space.

Space capabilities are a key element of the Armed Forces' multi-domain operations, acting as a multiplier of essential force.

The European Union as a whole is also taking measures to protect its space assets, defend its interests, deter irresponsible and potentially dangerous activities in space, and strengthen its strategic posture and autonomy. This Europe-wide involvement makes it necessary to jointly analyse the threats in and from space and to formulate an adequate response.

The EU Space Strategy for Security and Defence calls for the Single Intelligence Analysis Capability (SIAC) – operating under the High Representative for Foreign Affairs and Security Policy – together with the military and civilian intelligence services of the Member States, to enhance the Union's strategic understanding of space threats and counterspace operations. To this end, the High Representative, with the support of SIAC, has drafted the first annual classified analysis of the space threat landscape, which includes the evolution of anti-satellite capabilities.



Figure 15: COVE Space Operations and Surveillance Centre. Image: Air and Space Force

In addition to the above, and spurred by the European Commission and the EU Agency for the Space Programme (EUSPA), European companies in the space sector are supporting the formation of the EU Space Information Sharing and Analysis Centre (SIAC). This proactive security initiative, participated in by Spanish companies and, more recently, by the Spanish Space Agency, provides a platform to detect, identify and counter emerging threats. This network-based information-sharing platform promotes collaboration, awareness and best practices among private entities to ensure the security of our space systems, and that of the networks and services on which they depend.

The security of systems and services whose deployment, operation and use within the framework of the EU Space Programme may affect the security of the EU is regulated by Council Decision (CFSP) 2021/698. Within this framework, the EU External Action Service, under the leadership of the High Representative of the Union for Foreign and Security Policy, has developed the Space Threat Response Architecture (STRA) with detailed operating procedures, and conducts annual exercises. Exercise scenarios include the simulation of physical or cyber-attacks on terrestrial infrastructure, with the possible activation of Article 42.7 of the EU's mutual assistance policy.

One of the objectives of the EU Space Strategy for Security and Defence is to amend Decision 2021/698 to respond to all threats in the space domain that may affect the EU and its Member States. One of its working areas is that of civil-military cooperation in threat response mechanisms.

To this end, the 2025 NASS defines indicators evaluating the production and delivery of products enabling Space Situational Awareness (SSA) and Space Domain Awareness (SDA), the inclusion of military requirements in EU programmes, and Spain's participation in space threat response exercises and in EU projects focused on space operations and services. The indicators also reflect investment in strategic initiatives, the reduction of non-EU technological dependencies, growth in key EU programmes, and the strengthening of cybersecurity. Finally, they include attendance at governance meetings, membership of information-sharing networks, and progress in studies and testing for EU space programmes.

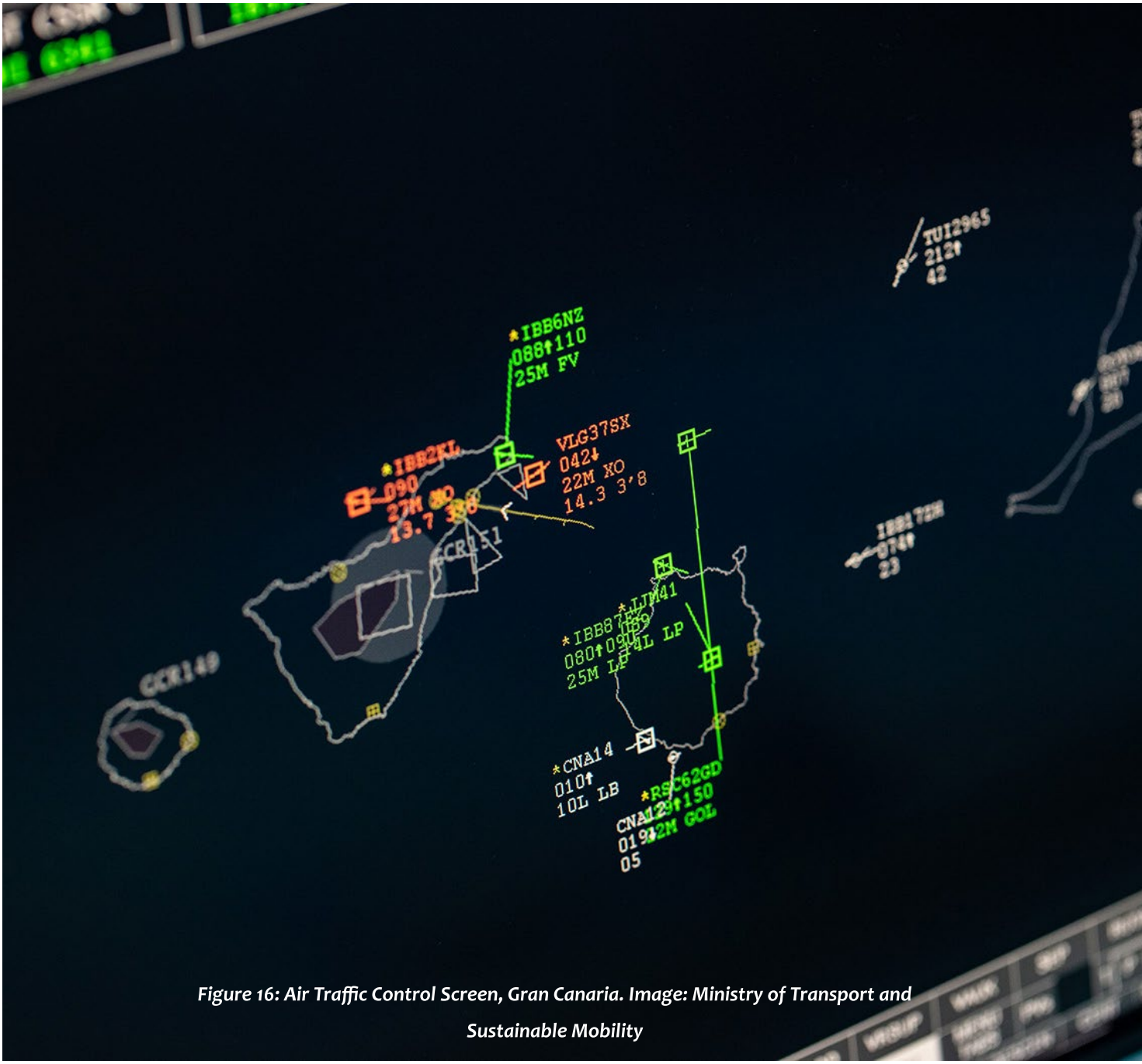


Figure 16: Air Traffic Control Screen, Gran Canaria. Image: Ministry of Transport and Sustainable Mobility



CHAPTER 3

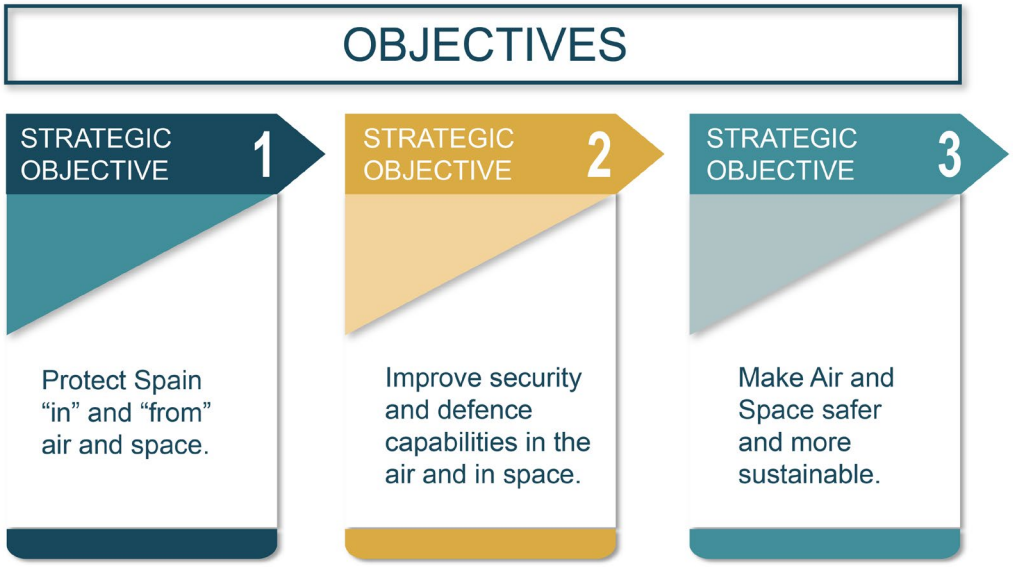
OBJECTIVES AND LINES OF ACTION

In an environment characterised by uncertainty and increasing geopolitical competition, it is vitally important to strengthen coordination and collaboration among public administrations, the private sector and international partners, in order to ensure the security and continuity of services in the air and space domains.

The Strategy identifies three strategic objectives:

- **Strategic Objective 1:** Protect Spain in and from air and space;
- **Strategic Objective 2:** Improve security and defence capabilities in the air and in space;
- **Strategic Objective 3:** Make air and space safer and more sustainable.

Figure 17: National Aerospace Security Strategy Objectives 2025. Image: DSN



Each of these strategic objectives is accompanied by corresponding lines of action, seeking to address threats, improve resilience and coordinate efforts, domestically and in the international context. The lines of action provide strategic political guidance to the relevant ministerial departments, which will decide on programmes within their respective areas of responsibility. Although several of the proposed lines of action contribute to achieving more than one strategic objective, for the sake of clarity, each is classified under a single heading.

The Annex to this Strategy presents a catalogue of the measures to be adopted to achieve the three Strategic Objectives.

Each measure is assigned a benchmark to facilitate evaluation of the degree of achievement towards the Objectives. To this end, the proposed lines of action and measures will be assessed every three years, by the National Aerospace Security Council, whose report will then be presented to the National Security Council.

The participation and cooperation of all levels of government, and especially of the Autonomous Communities with jurisdiction over security and industry, is considered essential to achieving the Strategic Objectives.

In both the lines of action and the measures included in the Annex there is an evident imbalance between the references to the air domain and those relating to space. This is due, on the one hand, to the novelty of considering the space domain separately, but also to the fact that the National Civil Aviation Security Programme and the State Civil Aviation Safety Programme had previously drawn up plans including lines of action, monitoring indicators and targets.

Finally, there is an important connection between these three Strategic Objectives and the five priority goals set out in the Industrial and Technological Plan for Security and Defence: 1) to modernise defence and deterrence capabilities; 2) to consolidate Spain's role as a reliable and prominent member of the European Union; 3) to promote innovation with respect to dual technologies; 4) to motivate the growth and internationalisation of the industrial fabric with respect to security and defence; 5) to enhance territorial cohesion by creating new industries and jobs throughout Spain.

Figure 18: 23rd Wing, F-5M and MQ-9 Predator B, Talavera la Real Air Base. Image: Air and Space Force



STRATEGIC OBJECTIVE 1:TO PROTECT SPAIN IN AND FROM AIR AND SPACE

This Objective includes developing the National Security System to address crisis situations whether they arise in or from air or space. Accordingly, the National Security System should prioritise:

- Development of the National Security Resource Catalogue.
- Development of resource preparation and deployment plans.
- Implementation of an Early Warning System based on indicators and alerts.

Within this Objective, the lines of action are focused on preserving the security and resilience of assets and services by intensifying surveillance, detection and tracking activities in both domains, protecting against cyberattacks and advancing our knowledge and understanding of the space environment.

The protection of national security and sovereignty requires effective collaboration among government entities, the private sector and international organisations to address any threat that may arise. From a national perspective, collaboration among the relevant ministries and agencies, the public sector and the private sector is key to enabling risk analysis, the planning of protection measures and the mitigation of adverse effects on critical entities and essential services.

We emphasise the importance of adequate cybersecurity awareness and the adoption of good practices and security update procedures. Moreover, an audit and certification scheme should be established, based on Directive (EU) 2022/2555 (NIS2), regarding the implementation of cybersecurity risk management measures and incident reporting obligations for aeronautical operators.

The aviation sector is extensively regulated, both by EU Directives and Regulations and by national legislation. However, the application of EU cybersecurity standards in the civil aviation sector introduces a high level of complexity, and airports, airlines

and service providers are all affected by a variety of regulatory frameworks. Failure to establish adequate coordination mechanisms might require a degree of compliance that could exceed their capabilities.

The first Strategic Objective calls for greater resilience of observation, communications, tracking and surveillance, control and air and space navigation systems, among others, to ensure the continuity of essential services through appropriate means, standards and procedures, taking into consideration the impact of advanced technologies.

Finally, several Lines of Action anticipate the development of higher airspace operations (HAO), the use of very low Earth orbits (VLEO) and congestion in low orbits, and advocate the provision of a regulatory framework to protect national interests.

In view of these considerations, to address Strategic Objective 1, the following Lines of Action are proposed:



Figure 19: Miura 1 Arenosillo Experimental Centre (INTA). Image: PLD Space

L.A. 1: Develop an appropriate regulatory framework for all space activities, structured in accordance with a Space Activities Act addressing security and defence interests in this domain.

L.A. 2: Promote the recruitment and training of personnel skilled in aerospace disciplines, and expand and improve the training offered in aerospace technologies and in aerospace law.

L.A. 3: Improve and refine civil-military coordination structures in air operations, via the flexible use of airspace, thus harmonising procedures, management and patterns of use. Ensure that Government aircraft can access and use airspace under appropriate conditions, especially in crisis situations. Adapt the architecture of communication, navigation and surveillance (CNS) systems and of air traffic management (ATM) systems to the needs of security and defence, including resilience measures and contingency plans.

L.A. 4: Strengthen the protection and resilience of the aerospace sector against cyberattacks through a comprehensive approach that encompasses all phases of the systems and facilities lifecycle, including specific cyber surveillance and cyber intelligence sharing measures.

L.A. 5: Develop measures to prevent, prepare for and respond to supply crises in the air and space domains in Spain, to ensure the availability and efficient use of resources under exceptional conditions, and to safeguard the industrial base that produces them.

L.A. 6: Strengthen coordination among all agencies regarding counter-drone powers and capabilities, ensuring the effective and comprehensive exchange of information. Promote collaboration and coordination with regional and local authorities, sharing and harmonising procedures for detecting potential criminal activity on these types of platforms.

L.A. 7: Develop, regulate and implement a Spanish system for space traffic management, in line with international recommendations.

L.A. 8: Develop policies and regulations for higher airspace operations, or operations at the edge of space, within Spanish airspace.

L.A. 9: Maintain secure Spanish and European communications and Earth observation capabilities, irrespective of potential market dominance by operators based outside the European Union.

L.A. 10: Maintain national and European positioning, navigation and precision timing capabilities. Establish and harmonise contingency protocols and procedures in the event of the loss or degradation of any of these services.

L.A. 11: Actively promote the use of a) the capabilities of the EU Space Programme; b) those derived from other international cooperation programmes or initiatives, as resources for preventing or responding to crisis or emergency situations. In particular, encourage Spanish public administrations, civil society and the armed forces to make use of the Copernicus programme, the Atlantic Constellation and the EU Earth Observation Governmental Service (EOGS) applications when they materialise.

L.A. 12: Ensure the availability of appropriate national aerospace capabilities to provide credible deterrence and an effective response across the full spectrum of crisis or conflict, ensuring their sustainability over time within a sufficient and stable budgetary framework.

L.A. 13: Review/update public health emergency preparedness and response plans in the aviation sector, taking into account that the State Preparedness and Response Plan for Major Threats to Public Health will establish the mechanisms for declaring emergencies of national importance, the requisite governance mechanisms, and the capabilities and resources necessary to ensure timely preparation and response to these emergencies.

STRATEGIC OBJECTIVE 2: IMPROVE SECURITY AND DEFENCE CAPABILITIES IN THE AIR AND IN SPACE

In a constantly evolving landscape of global threats, it is imperative to enhance air and space security capabilities, ensuring availability and operational readiness, to enable the continued and sustainable use of the services and systems that rely on these assets.

It is also of vital importance to develop new surveillance, protection and threat response capabilities that incorporate the most advanced technologies into a multi-layered, in-depth defence. To achieve this, we must continue the development by Spanish industries of air defence technologies, aircraft and weapons, as well as their participation in multinational programmes, within NATO and the EU.

Access to space is essential. Therefore, the development of propulsion and launch reuse technologies should be a national and European priority.

The protection and resilience of aerospace infrastructure requires the incorporation of new physical, electromagnetic and cybersecurity protection measures, combined with the integration of emerging technologies and the application of disruptive technologies, such as AI for new generations of sensors.

It is also essential to continue developing the dual use of EU Space Programme components (Galileo/EGNOS, Copernicus, IRIS2, GOVSATCOM and SSA) and to modernise the technological capabilities that complement or enhance them, in order to achieve space situational awareness and real-time warning capabilities. National systems, both civil and military, must be strengthened in order to enhance our ability to detect and classify threats in or from space in the shortest possible time.

The development of national capabilities, preferably dual, in the surveillance and tracking of the space domain is another area of priority attention. This should be conducted in coordination with partner and allied countries in our region, either bilaterally or through participation in programmes of the European Space Agency, NATO and the European Union.

These capabilities include improving the ability to detect and track space objects, as a fundamental aspect of space security and defence, thus upgrading space traffic management capabilities and achieving more precise space environment awareness.

In view of these considerations, it is of great importance to cooperate with the private sector, incorporating their capabilities into space security policies, and promoting a culture of space security and defence at all levels. Strategic communication is essential to raising public awareness about the importance of space for the security and well-being of the Spanish population.

To achieve this Objective, the following Lines of Action should be undertaken to improve aerospace infrastructure, promote strategic autonomy, boost scientific research and integrate new technologies, with a strong focus on AI.



Figure 20: 21st Air Surveillance Squadron, Pico de las Nieves, Gran Canaria. Image: Spanish Air and Space Force

L.A. 14: Improve space access capabilities through national launcher programmes and participation in European equivalents. Promote the establishment and use of infrastructure for space access from Spanish territory, possibly including the authorisation of spaceports on Spanish territory.

L.A. 15: Improve air defence capabilities by increasing the number and performance of aircraft and weapons, both airborne and ground-based. Improve the capabilities of national air and space surveillance and control systems, with a significant increase in sensors and their protection.

L.A. 16: Strengthen and increase national counter-drone capabilities, both for public safety and for security and defence purposes, through the advanced integration of data from various sensors and the application of AI techniques to optimise analysis and operational response.

L.A. 17: Develop and implement AI within civil and military air and space surveillance and control systems, taking into account that Big Data and AI techniques enable the integration of information from all types of sensors, with virtually no limitations on number or type.

L.A. 18: Further develop satellite-based quantum key distribution technologies to ensure the security of government communications and strategic assets.

L.A. 19: Improve Spanish systems for space operations surveillance (MESPA/COVE; AEE/S3TOC) and implement coordination mechanisms with air traffic services. Create and maintain an updated national catalogue of space objects, with the highest standards of information quality and security.

L.A. 20: Promote scientific and technological knowledge and competitiveness in systems and services, for the generation of spatial situational awareness capabilities.

L.A. 21: Develop and implement the standards and technical infrastructure necessary to facilitate government and private-sector space operations.

L.A. 22: Establish a Spanish space meteorology service, sufficient to satisfy the nation's basic needs in this respect, and create an alert system for civil protection and for State defence and security services.

L.A. 23: Develop the regulatory, inspection and oversight capabilities of public administrations to ensure that operators and service providers in the air and space domains are properly equipped to manage cybersecurity.

L.A. 24: Promote the development of dual-use space systems enabling shared use by security and defence forces, public administrations and non-governmental users. Particular attention should be paid to the requirements for supporting emergency and disaster management.

L.A. 25: Promote the development of military space systems capable of protecting satellite systems of national interest.

STRATEGIC OBJECTIVE 3: MAKE AIR AND SPACE SAFER AND MORE SUSTAINABLE

This Objective reflects Spain's commitment to security and sustainability in the aerospace sector, involving both national and international efforts. Accordingly, Spain will strengthen its participation in international initiatives under the EU and NATO Common Security and Defence Policy, enhancing cooperation in the surveillance and protection of aerospace systems. In this regard, and together with our allies, it will be essential to strengthen NATO's integrated air and missile defence (IAMD) capability, both within and outside our national territory.

In addition to the above, Spanish participation and investment in international aerospace security organisations and forums should be promoted. Furthermore, bilateral and multilateral aerospace security agreements will be sought for mutual support in crises, especially with countries in Europe and elsewhere that share the same security principles.

It is of priority importance to enhance the exchange of space situational awareness information with international organisations and to ensure interoperability with NATO and EU command and control systems, particularly those associated with space surveillance.

Measures must be adopted to ensure the sustainable use of space, together with the continued incorporation of new sensors and satellites that improve the effectiveness of international protocols for predicting and responding to adverse weather events.

Achieving and maintaining the sustainability of operations in the air and space domains requires further technological research into sustainable propellants and fuels to provide strategic autonomy and prevent vulnerability in the supply chain.

Finally, this Objective requires us to assess the economic impacts of meteorology and space debris, among other dangers, by developing impact maps and adopting protective measures.

L.A. 26: Continue to share analysis with our EU and NATO partners and allies and adopt a common response to threats in the space domain. In particular, develop national responsibilities within the EU Space Threat Response Architecture (STRA) and actively participate in the EU Space Information Sharing and Analysis Centre (ISAC).

L.A. 27: Strengthen international cooperation and coordination within the European Union and NATO to enable an appropriate response to any crisis (including health crises and armed conflicts) that could impact on air traffic and airport operations.

L.A. 28: Develop international protocols for action and warnings of space alerts to enable coordinated action, especially within the European Union, in the event of threats in or from space. Particular attention should be paid to coordinated action at the European level on measures that may affect air traffic and operational safety.

L.A. 29: Develop national action protocols to enable coordinated action by public authorities in any crisis that could impact on air traffic and airport operations.

L.A. 30: Develop an open ecosystem of spatial data spaces to enable the exploitation of AI capabilities and access to a wide range of data and services, following the models of the Copernicus Data Space Ecosystem and the Spanish Space Data Infrastructure.

L.A. 31: Mitigate the effects of space debris on space operations and activities and minimise the generation of new waste. Promote the preservation of dark and quiet skies, which have instrumental value for science and are closely linked to the long- and medium-term sustainability of space activities.

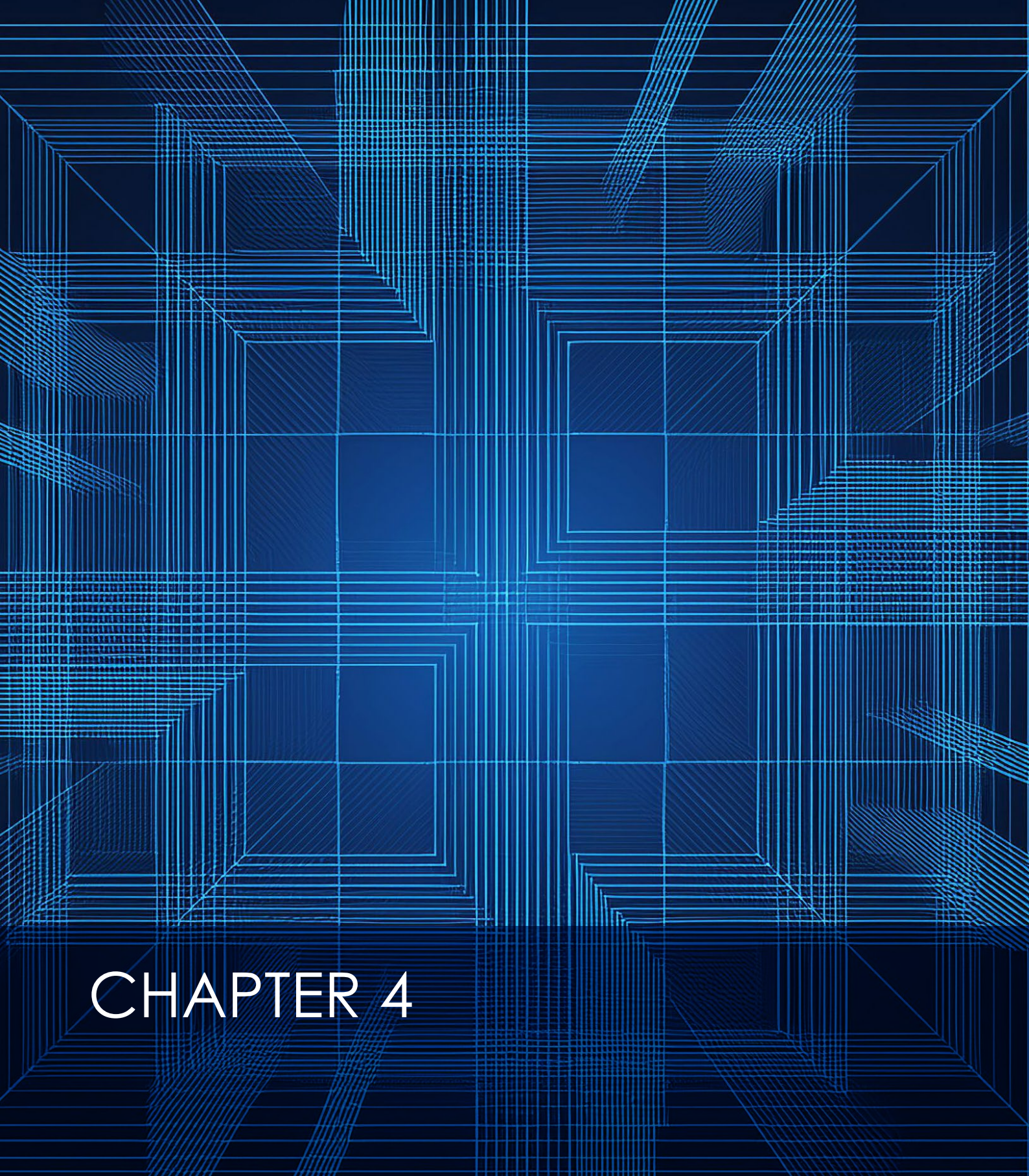
L.A. 32: Improve interoperability and data exchange to develop space situational awareness. Conclude information-sharing agreements with other international stakeholders in this field, in addition to the EU and the ESA. Facilitate access to basic space information services.

L.A. 33: Develop international standards and best practices for space operations to facilitate space traffic management, participating in forums where these standards are being developed. Coordinate the resolution of potential sources of radio interference, whether intentional or unintentional, with the responsible organisations, or through bilateral agreements with key nations, or through procedures established by the International Telecommunication Union.

L.A. 34: Develop new decarbonisation technologies in the aerospace sector, particularly sustainable aviation fuel and hydrogen, and develop high-energy-density batteries, which are essential for the development of electric vertical take-off and landing (eVTOL) aviation.



Figure 21: Seville Solar Plant, Paz-1 image. Image: HISDESAT



CHAPTER 4

AGENCIES WITHIN THE NATIONAL SECURITY SYSTEM WITH FUNCTIONS RELATED TO AEROSPACE SECURITY

The nature and structure of aerospace security agencies within the National Security System are determined by Act 36/2015, on National Security, by Order PCM/218/2020, which created and regulates the National Aerospace Security Council, and by Order PCM/1067/2022, which approved the Space Alerts Protocol. This structure includes:

A. The National Security Council (CSN).

B. The National Aerospace Security Council (CNSA).

C. The Situation Committee, a single committee for the entire National Security System in crisis situations.

D. The Coordination Cell

The primary function of this structure is to effectively manage crises in the aerospace domain. These agencies assess the factors and situations that may affect National Security, gather and analyse information to make the necessary decisions to direct and coordinate the response to crisis situations, identify needs and propose planning and coordination measures with the relevant public administrations in the National Security System.

THE ORGANISATION OF AEROSPACE SECURITY

A. THE NATIONAL SECURITY COUNCIL

The National Security Council (CSN) exercises the functions assigned to it by Act 36/2015, on National Security. The CSN is the Government's Delegate Commission for National Security and assists the President of the Government in determining National Security policy and the National Security system.

B. THE NATIONAL AEROSPACE SECURITY COUNCIL

The creation of a National Aerospace Security Council (CNSA) was one of the measures required by the 2019 National Aerospace Security Strategy.

The CNSA was created and regulated by Order PCM/218/2020, as a specialised group in the field of aerospace security and as a support body for the National Security Council within the framework of the National Security System. The creation of the Spanish Space Agency led to the composition and representation of the CNSA being modified, by Order PCM/549/2023.

The CNSA is headed by the Chief of Defence (JEMAD). Its composition reflects the transversal nature of the air and space domains and enables the incorporation of key public-sector stakeholders in the field of aerospace security, and, when appropriate due to issues of interest or jurisdiction, representatives of the Autonomous Communities and Cities. It also includes representatives from other entities from the institutional

public sector and from the private sector, together with individuals in their capacity as experts whose contributions are deemed relevant.

To support the Situation Committee in the area of aerospace security, the CNSA performs the following functions: assessing the risks associated with threats; analysing potential crisis scenarios and their evolution, especially those likely to lead to a situation of interest to National Security; developing and maintaining updated response plans; and formulating guidelines for conducting crisis management exercises and evaluating their results. At all times, the CNSA coordinates its activities with the directly competent bodies and authorities and with the Situation Committee.

The Department of National Security serves as the Technical Secretariat and a permanent working body of the CNSA.

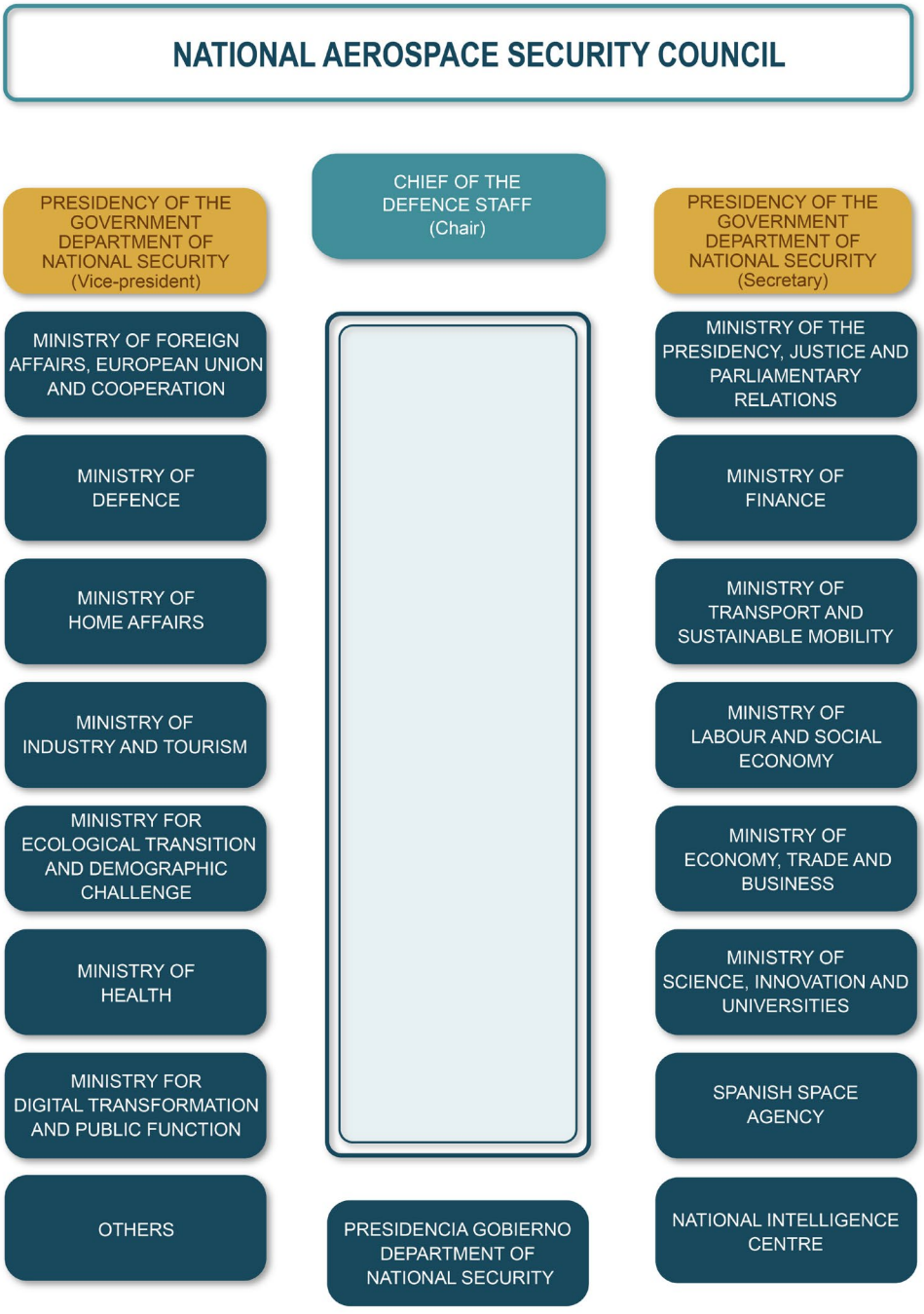


Figure 22: Composition of the National Aerospace Security Council. Image: DSN

C. THE SITUATION COMMITTEE

The Situation Committee is convened to manage crisis situations in the field of aerospace security under the direction of the CSN, whenever the transversality, dimension and/or impact of such situations exceeds the effective response capacity of the normal mechanisms, always respecting the powers assigned to corresponding agencies and institutions and for the purposes of ensuring an immediate, coordinated and effective response via a single body to conduct the strategic political management of the crisis.

The Situation Committee and the CNSA act in a complementary manner, each within its own area of expertise, but under the single strategic and political direction of the CSN, chaired by the President of the Government.

The Department of National Security supports the Situation Committee to ensure effective liaison with the operational centres involved and to provide an appropriate response in crisis situations, facilitating monitoring, control and the transmission of decisions. To this end, it may use the special communications network of the Office of the President of the Government.

D. COORDINATION CELL

The Department of National Security, in accordance with the provisions of Article 20.4 of Act 36/2015, maintains permanent liaison and coordination mechanisms with the agencies of all areas of public administration that are necessary for the National Security System to exercise its functions and fulfill its objectives, uninterruptedly and without prejudice to the functions corresponding to the Situation Committee in matters of crisis management.

To effectively carry out its support functions for the Situation Committee, the Department of National Security may be reinforced by specialised personnel from the relevant ministerial departments or agencies, who will form a specific Coordination Cell for Aerospace Security. This Cell is regulated by the stipulations for working groups referred to in Point Nine of Order PMC/32/2018.

In this regard, Order PCM/1067/2022 approved the Space Alerts Protocol, which provides for the activation of a Coordination Cell as a preventive instrument among the CSN's

coordination and support bodies, in accordance with the provisions of Article 26.2 of Act 36/2015 on National Security. If the preventive activation of the National Security System takes place due to a contingency contemplated in the Space Alerts Protocol or, where appropriate, the declaration of a situation of interest to National Security by the President of the Government, the Coordination Cell will be immediately activated by the Director of the Department of National Security, and will begin its technical advisory work in the premises of the Situation Centre of the Department of National Security, providing support for the Specialised Situation Committee.

The Coordination Cell integrates and analyses the information obtained, facilitating a liaison and coordination mechanism among the ministries and agencies involved. Each ministry and agency carries out the actions that correspond to it within its jurisdiction.

The Coordination Cell, as provided for in the Space Alert Protocol, which was initially limited to uncontrolled re-entries into the Earth's atmosphere, has proven to be an effective tool that provides the necessary responsiveness to the National Security System. It is therefore considered the appropriate instrument to support the management of other scenarios related to threats in or from space that affect national security, and in particular to facilitate cooperation with the EU Space Threat Response Architecture.



Figure 23: EF2000 Eurofighter Typhoon (C.16). Image: Air and Space Force

ANNEX

MEASURES AIMED AT ACHIEVING THE THREE STRATEGIC OBJECTIVES

STRATEGIC OBJECTIVE 1:

PROTECT SPAIN IN AND FROM AIR AND SPACE

ACTION 1: ENSURE SECURITY IN THE AEROSPACE SUPPLY CHAIN.

ACTION 2: IMPROVE PROTECTION AND RESILIENCE TO CYBERATTACKS FROM AEROSPACE.

ACTION 3: IMPROVE THE SECURITY OF CRITICAL AERONAUTICAL AND SPACE INFRASTRUCTURE, THAT OF CRITICAL, ESSENTIAL AND IMPORTANT AERONAUTICAL AND SPACE ENTITIES, AND THAT OF THE CAPABILITIES AND SERVICES THEY PROVIDE.

STRATEGIC OBJECTIVE 2:

IMPROVE SECURITY AND DEFENCE CAPABILITIES IN THE AIR AND IN SPACE

ACTION 4: PROMOTE NATIONAL AUTONOMY IN ACCESS TO SPACE.

ACTION 5: HIGHLIGHT THE SECURITY DIMENSION IN SPACE PROGRAMMES, BOTH NATIONAL AND IN THE CONTEXT OF THE EU.

ACTION 6: STRENGTHEN NATIONAL SPACE SITUATIONAL AWARENESS CAPABILITIES, AND REINFORCE THE EXCHANGE AND INTEROPERABILITY OF SPACE INFORMATION.

ACTION 7: BOOST SPACE DOMAIN AWARENESS CAPABILITIES.

ACTION 8: PROMOTE THE DEVELOPMENT OF DISRUPTIVE TECHNOLOGIES IN THE DOMAINS OF AIR AND SPACE.

ACTION 9: STRENGTHEN THE COMPETITIVENESS OF THE SPACE INDUSTRY.

ACTION 10: STRENGTHEN THE COMPETITIVENESS OF THE AERONAUTICAL INDUSTRY.

ACTION 11: STRENGTHEN RELATIONS BETWEEN PUBLIC ADMINISTRATIONS, THE PRIVATE SECTOR AND R&D&I CENTRESi.

STRATEGIC OBJECTIVE 3:

MAKE AIR AND SPACE SAFER AND MORE SUSTAINABLE

ACTION 12: PROMOTE SPACE AS A COMMON GOOD, A FACTOR OF INTEREST TO NATIONAL SECURITY AND A DRIVER OF PROGRESS.

ACTION 13: CONTRIBUTE TO THE SPACE SECURITY ACTIVITIES OF THE INTERNATIONAL ORGANISATIONS TO WHICH SPAIN BELONGS.

ACTION 14: STRENGTHEN COOPERATIVE SECURITY IN BILATERAL AGREEMENTS OF STRATEGIC INTEREST TO SPAIN.

ACTION 15: CONTRIBUTE TO THE SUSTAINABLE USE OF SPACE ORBITS.

ACTION 16: DEVELOP INTERNATIONAL PROTOCOLS FOR THE USE AND TRANSMISSION OF SPACE ALERTS.

MEASURES TO ACHIEVE STRATEGIC OBJECTIVE 1

PROTECT SPAIN IN AND FROM AIR AND SPACE

ACTION 1: ENSURE SECURITY IN THE AEROSPACE SUPPLY CHAIN

Industrial policy is an essential element of national security, and ensuring security in the supply chain for the aerospace sector is a key component of both questions.

The competent authorities will develop the necessary capabilities and appropriate control procedures to ensure the security of the industry's supply chain, as this is a highly significant risk factor.

For all programmes developed to acquire aerospace capabilities, the analysis of supply chain security risks will be optimised, taking a proactive approach from the outset.

All necessary protective measures will be taken, leveraging new capabilities offered by technologies such as artificial intelligence.

In view of these considerations, it will be necessary to develop a national or European auxiliary industry to provide the necessary components for advances in space, in all areas of activity, especially those most vulnerable to intentional attacks. Declarations of the source of component supplies will be required in pre-programme documentation and risk analyses.

Special attention will be paid to protecting industrial information related to products, services, procedures and/or strategies whose confidentiality represents added value or a competitive advantage for Spanish and European industry.

The capabilities of the Autonomous Communities and Cities with responsibilities in industry and security will be integrated to ensure collaborative effectiveness.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Foreign Affairs, European Union and Cooperation; Ministry of Defence; Ministry of Home Affairs; Ministry of Industry and Tourism; Ministry of Science, Innovation and Universities; Ministry for the Ecological Transition and the Demographic Challenge; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency; Autonomous Communities and Cities.

MEASURES OF ACHIEVEMENT

The initiatives undertaken by each agency involved, with the participation of private entities in matters of supply chain security (especially in the analysis of associated risks).

Incidents that might impact on the security of industrial information.

ACTION 2: IMPROVE PROTECTION AND RESILIENCE TO CYBERATTACKS FROM AEROSPACE

Cyberattack prevention, detection, monitoring and response capabilities will be strengthened by activating the plans outlined in the National Cybersecurity Strategy and providing the necessary resources for this purpose.

Action will be taken to ensure that the information and communications systems utilised in this area conform to national and European standards and, where appropriate, to other international requirements.

The use of reliable, certified solutions, products, systems and services for sensitive networks will be promoted. Security requirements in this respect must be stipulated at the initial stage of space systems design.

Cybersecurity awareness will be raised among key national players in the aerospace sector, especially the companies involved, via the adoption of appropriate procedures and best practices, including those related to system security updates.

Cybersecurity criteria will be incorporated both into the documents that define requirements and feasibility for processes of aerospace capability acquisition, and also into the maintenance activities carried out on critical platforms and facilities related to the sector.

Cyberthreat intelligence will be utilised to provide predictive and strategic value, as an indispensable complement to physical and logical security measures. When appropriate, counterintelligence measures will be applied to help counter cyberthreats in the aerospace sector.

Once the Strategy is approved, the cybersecurity requirements of the Regulation on the Safety, Resilience and Sustainability of Space Activities (the EU Space Act), which acts as a *lex specialis* to the NIS2 Directive, will be incorporated.

The Autonomous Communities and Cities will collaborate as necessary, to the extent of their capabilities.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Defence; Ministry of Home Affairs; Ministry of Transport and Sustainable Mobility; Ministry of Science, Innovation and Universities; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency; State Agency for Aviation Safety; State Agency for Digital Administration; Autonomous Communities and Cities.

MEASURES OF ACHIEVEMENT

A catalogue will be created of the main operators and their supply chains in the sector, detailing the awareness-raising actions to be carried out.

A minimum level of resilience, common to all operators, will be defined and applied, with the corresponding criteria of proportionality.

A cybersecurity operations centre dedicated to the air and space sectors will be created to enable the systematic reporting of security incidents; this centre's function will be integrated with other initiatives proposed in the European Union.

Measures for cybersecurity oversight in the aerospace sector will be defined and adopted.

Cybersecurity incident response protocols for the aerospace sector will be defined and adopted.

ACTION 3: IMPROVE THE SECURITY OF CRITICAL AERONAUTICAL AND SPACE INFRASTRUCTURE, THAT OF CRITICAL, ESSENTIAL AND IMPORTANT AERONAUTICAL AND SPACE ENTITIES, AND THAT OF THE CAPABILITIES AND SERVICES THEY PROVIDE

With the participation of the relevant Ministries and Agencies and the collaboration of the private sector, a risk analysis will be prepared of existing land-based infrastructure, based on its level of vulnerability, as a necessary preliminary to planning measures for its protection.

A catalogue will be made of cybersecurity measures required under the NIS2 Directive and of the regulations that transpose it. In addition, an audit and certification scheme for compliance with these measures will be implemented, in accordance with the Directive.

A similar catalogue will be made of measures required under the Critical Entities Resilience (CER) Directive and of the regulations that transpose it.

In the space domain, once approved, the regulations for the protection of critical space infrastructure will be aligned with the Regulation on the safety, resilience and sustainability of space activities (the EU Space Act).

The designated agencies will coordinate the execution of the process in collaboration with the relevant Ministry.

The Autonomous Communities and Cities will collaborate as necessary, to the extent of their capabilities.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Defence; Ministry of Home Affairs; Ministry of Transport and Sustainable Mobility; Ministry of Science, Innovation, and Universities; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency; State Aviation Safety Agency; Autonomous Communities and Cities.

MEASURES OF ACHIEVEMENT

1. The specific cybersecurity measures adopted for the protection of infrastructure of interest in the aerospace field, in accordance with the NIS2 Directive, the resulting implementing acts, the provisions of the regulation that transposes this Directive into the Spanish legal system, and its regulatory development.

2. The specific security measures adopted for the protection of infrastructure of interest in the aerospace field as established in the CER Directive and the implementing acts derived from it.

3. An aerospace cybersecurity report, determining compliance with established cybersecurity measures.

4. Ongoing reports detailing the surface area of the above infrastructures, reflecting the vulnerabilities detected.

5. Cybersecurity requirements established for critical digital products used in the aerospace sector, together with taxonomies for these products in the Information and Communications Technology Security product catalogue.

6. The Sectoral Strategic Plans for aeronautical and space infrastructure will be reviewed and drafted by a Working Group within the National Centre for the Protection of Critical Infrastructure (CNPIC), in accordance with Royal Decree 704/2011, which approved the Regulation for the protection of critical infrastructure.

7. A catalogue of national companies working with dual-use technologies in aeronautical and space programmes.

MEASURES TO ACHIEVE STRATEGIC OBJECTIVE 2

IMPROVE SECURITY AND DEFENCE CAPABILITIES IN THE AIR AND IN SPACE

ACTION 4: PROMOTE NATIONAL AUTONOMY IN ACCESS TO SPACE

The strategic nature of the space domain necessitates the ability to autonomously launch space assets (rockets and satellites).

Possession of this ability facilitates decision-making, obviating the need to depend on other space agencies or countries. In other words, it provides greater control, and therefore security, over missions and objectives of interest.

Autonomy in launchers, in locations and in facilities to perform launches from national territory will also allow us to attract investment and promote experimentation and the development of new technologies and initiatives related to space activity, thus creating jobs in the sector and helping transform the economy.

ORGANISMOS IMPLICADOS

Ministry of Defence; Ministry of Industry and Tourism; Ministry of Science, Innovation, and Universities; Ministry for Digital Transformation and Public Service; Spanish Space Agency.

MEASURES OF ACHIEVEMENT

Each Ministerial department will provide information on the investment planned and on the activities implemented to boost national capacity for spaceport and launch infrastructure.

ACTION 5: HIGHLIGHT THE SECURITY DIMENSION IN SPACE PROGRAMMES, BOTH NATIONAL AND IN THE CONTEXT OF THE EU

The security dimension of national and EU space programmes will be promoted in various forums.

The design, construction, operation and maintenance of satellites and satellite-based services will be conducted with a view to strengthening and sustaining Spain's strategic space autonomy.

EU financing mechanisms will be utilised to strengthen, consolidate and enhance the aerospace industrial base, thus contributing to the nation's economy and security.

Spain will promote and actively participate in the development of a Space Domain component within the EU Space Programme, generating specific plans and programmes to heighten security and defence capabilities.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Foreign Affairs, European Union and Cooperation; Ministry of Defence; Ministry of Transport and Sustainable Mobility; Ministry of Industry and Tourism; Ministry for the Ecological Transition and the Demographic Challenge; Ministry of Science, Innovation and Universities; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency.

MEASURES OF ACHIEVEMENT

The security component in national space programmes.

EU Space Programme security-related facilities located in Spain.

Projects in which Spain participates, stating the number of monitoring and control facilities deployed in national territory.

Space Domain Awareness projects with Spanish participation; number of space surveillance sensors and systems deployed in Spain.

ACTION 6: STRENGTHEN NATIONAL SPACE SITUATIONAL AWARENESS CAPABILITIES, AND REINFORCE THE EXCHANGE AND INTEROPERABILITY OF SPACE INFORMATION

National capabilities in space surveillance and tracking (SST), space weather events (SWE), and near-Earth object (NEO) detection will be enhanced by incorporating emerging technologies in ground- and space-based sensors, intelligent systems and redundant communications with artificial intelligence. The enhancement and modernisation of these capabilities will facilitate effective responses within an increasingly dynamic and saturated space environment, ensuring early warning and real-time analysis.

The comprehensive and coordinated management of space situational awareness will be promoted, enhancing the interoperability of civilian and military operational centres through automatic and systematic information exchange processes (the "single data" concept), thus ensuring data consistency. To this end, regular meetings will be held among national officials and international cooperation with partners and allies will be fostered.

Risk scenarios will be developed to assess threats, and a joint early warning system based on objective indicators will be strengthened.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Defence; Ministry of Transport and Sustainable Mobility; Ministry of Science, Innovation, and Universities; Ministry for the Ecological Transition and the Demographic Challenge; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency; AEMET; National Geographic Institute.

MEASURES OF ACHIEVEMENT

The percentage of near-Earth space surveillance coverage performed by Spanish systems.

The national resources and infrastructure available for monitoring space phenomena (geomagnetic field, ionosphere, cosmic radiation, NEO) and for the detection and monitoring of potential threats in this domain.

Inter-ministerial cooperation protocols.

International agreements for interoperability and information exchange.

ACTION 7: BOOST SPACE DOMAIN AWARENESS CAPABILITIES

Current space surveillance capabilities will be consolidated and then strengthened, thus achieving optimum Space Domain Awareness (SDA), facilitating the generation of alerts, the anticipation of threats and dangers and, where appropriate, the attribution of actions, via the analysis and integration of information obtained, through internal means and/or collaboration with other parties.

Space intelligence and asset protection capabilities will be developed to maintain protection of critical space systems against electronic warfare, disruption, disablement, neutralisation or destruction.

To this end, plans will be promoted to obtain space capabilities for the Armed Forces.

The Ministry of Defence's SDA capabilities, with special regard to Space Command, will be enhanced, promoting secure interconnection between systems of different classification levels.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Defence; Ministry of Science, Innovation and Universities; Ministry for the Ecological Transition and the Demographic Challenge; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency.

MEASURES OF ACHIEVEMENT

The acquisition of capabilities according to a plan by which the means to generate SDA will be obtained.

The creation of intelligence databases that include the characteristics, regulations, limitations, vulnerabilities, intended threats, capabilities, behaviours and other information regarding other relevant forces in space.

The monitoring and reporting of events that may generate or have generated space alerts and their implications for security and defence.

Establishment of the structures necessary for the generation of SDA and the Recognised Space Picture (RSP), establishing defensive mechanisms and procedures and determining response options.

Bilateral agreements or agreements with international organisations or agencies that promote interoperability and information exchange.

ACTION 8: PROMOTE THE DEVELOPMENT OF DISRUPTIVE TECHNOLOGIES IN THE DOMAINS OF AIR AND SPACE

In the current era, marked by an increasing number and variety of threats in the domains of air and space, they can be simplified into three main axes: physical, electromagnetic and cyber. In this context, it is necessary to achieve the greatest possible degree of threat anticipation, via the development of new technologies applied to space capabilities.

By promoting innovation and technological and industrial foresight, achieved by means of appropriate knowledge and research centres, it will be possible to acquire technological and industrial advantages in certain knowledge niches, which will be key to improving security and protection measures for space activity.

In one aspect of the above, technologies will be developed to improve counter-drone capabilities, both in airspace surveillance and control systems and in drone interdiction at all flight levels and speeds.

Another priority is the need to develop and employ air and space domain capabilities to predict and respond to crisis situations caused by natural disasters, using artificial intelligence.

The Autonomous Communities and Cities will collaborate as necessary, to the extent of their capabilities. .

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Defence; Ministry of Home Affairs; Ministry of Transport and Sustainable Mobility; Ministry of Industry and Tourism; Ministry of Science, Innovation and Universities; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency; Autonomous Communities and Cities.

MEASURES OF ACHIEVEMENT

Projects and investments to promote disruptive space technologies.

Projects and investments to promote disruptive anti-drone technologies.

Research projects in disruptive aerospace technologies at universities and research centres.

Crisis modelling and simulation capabilities.

ACTION 9: STRENGTHEN THE COMPETITIVENESS OF THE SPACE INDUSTRY

The growth of the space industry will be promoted to create a solid technological and industrial base. This will enable national objectives to be met and facilitate participation in international programmes within the EU, the European Defence Agency, the European Space Agency (ESA) and NATO, among others.

At the national level, developments coherent with the National Plans promoted and/or coordinated by the ESA and the Ministry of Defence will be supported.

These developments should be aimed at strengthening the competitiveness and autonomy of the space sector, reducing our dependence on third countries and promoting technological solutions, entrepreneurship, innovation, digitalisation, sustainability and the circular economy. The overall goal is to establish a competitive national space industry that is climate neutral.

The intellectual and industrial property of strategic developments will be protected, dual technologies will be promoted and Spanish industry will be encouraged to participate in international programmes on a balanced basis. Success in these areas will generate the benefits inherent in collaborations, while ensuring a controlled exchange of knowledge and experience.

In all of the above, the aim is to achieve a high level of technological absorption, whilst ensuring the security of developments and leveraging the synergies generated and optimising investment with other stakeholders in shared capabilities.

The security dimension of innovation, entrepreneurship, basic research and technological development will also be promoted by identifying advances with high strategic value and, where appropriate, by safeguarding our technological and industrial heritage. This will ensure that Spain and Spanish industry are adequately protected, without limiting participation in the international deployment of Spanish companies' high-value-added capabilities.

The Autonomous Communities and Cities will collaborate as necessary, to the extent of their capabilities.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Defence; Ministry of Home Affairs; Ministry of Transport and Sustainable Mobility; Ministry of Industry and Tourism; Ministry of Science, Innovation and Universities; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency; Autonomous Communities and Cities.

MEASURES OF ACHIEVEMENT

Public and private investment in R&D&I projects for the space sector.

Capacity-building strategic technology projects within national and international programmes, determining the financial allocation made and the technological readiness level (TRL) sought and achieved.

Measures for the protection of intellectual property.

ACTION 10: STRENGTHEN THE COMPETITIVENESS OF THE AERONAUTICAL INDUSTRY

Strengthening the aeronautical industry in Spain is a strategic task, as this sector has a significant economic, technological and social impact. Spain must align its policies and strategies with international standards on the reduction of greenhouse gas emissions to ensure the sector's global competitiveness, promoting research into new sustainable fuels, electric or hybrid aircraft and cleaner manufacturing processes.

The aeronautical sector is one of the most technologically advanced, and Spain has the opportunity to become a leader in innovation within global aviation. By strengthening the aeronautical industry, cutting-edge technologies can be generated and transferred to other industrial sectors, contributing to the modernisation of the economy as a whole.

Grants will be offered for research and development projects to produce new aircraft and equipment, or improvements to current manufacturing processes, with specific, long-term programmes that support investment in the aeronautical sector and ensure the greatest possible strategic autonomy, promoting the development of domestic technologies and reducing dependence on other countries for the supply of critical components and the acquisition of essential aeronautical capabilities.

The integration of capabilities across the entire value chain will be fostered, particularly between industry and research organisations, creating synergies in key areas such as artificial intelligence, uncrewed aircraft, additive manufacturing and green aviation.

In all these areas, the overall aim is to enhance Spain's industrial capacity in systems and equipment in fields beyond that of aerostructures and to maintain our leadership in priority technologies such as composite materials.

The Autonomous Communities and Cities will collaborate as necessary, to the extent of their capabilities.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Defence; Ministry of Home Affairs; Ministry of Industry and Tourism; Ministry of Science, Innovation and Universities; Spanish Space Agency; Autonomous Communities and Cities.

MEASURES OF ACHIEVEMENT

Public and private investment in R&D&I projects for the aeronautic sector.

Capacity-building strategic technology projects within national and international programmes, determining the financial allocation made and the technological readiness level (TLR) sought and achieved.

Measures for the protection of intellectual property.

ACTION 11: STRENGTHEN RELATIONS BETWEEN PUBLIC ADMINISTRATIONS, THE PRIVATE SECTOR AND R&D&I CENTRES

If a security threat or event occurs, the orientation and evaluation of possible response options must be initiated. To minimise the impact of any threat, appropriate structures and protocols must be employed to minimise reaction time and expedite the response. Moreover, these mechanisms should enable the participation of public authorities, with appropriate coordination. Finally, powers should be delegated when so required, thus decentralising the response made.

Formal relations with the private sector will be strengthened to obtain their essential contribution to aerospace safety objectives, without prejudice to the obligations and responsibilities established by current legislation. In this, our aim is to establish an appropriate framework for cooperation between the public and private sectors, involving dialogue and the strategic analysis of shared risks and opportunities, to

inform and guide the comprehensive and efficient action of all national stakeholders in this respect.

Active communication between public administrations and the space industry will be encouraged to channel initiatives such as regulatory improvements and joint participation in matters of mutual interest.

The participation of private entities in exercises: in a variety of scenarios, different procedures and protocols will be practised to improve mutual support in response to situations that may arise.

The Autonomous Communities and Cities will collaborate as necessary, to the extent of their capabilities.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Defence; Ministry of Home Affairs; Ministry of Transport and Sustainable Mobility; Ministry of Science, Innovation, and Universities; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency; State Aviation Safety Agency; Autonomous Communities and Cities.

MEASURES OF ACHIEVEMENT

The initiatives undertaken by relevant Ministries and Agencies with the participation of private entities in matters of joint interest.

Exercises performed.

MAKE AIR AND SPACE SAFER AND MORE SUSTAINABLE

ACTION 12: PROMOTE SPACE AS A COMMON GOOD, A FACTOR OF INTEREST TO NATIONAL SECURITY AND A DRIVER OF PROGRESS

All parties involved will be urged to help enhance knowledge and understanding of the value that security in and from space offers for the overall security and well-being of the country and its citizens, and to foster public interest in this new field.

Strategic communication campaigns will be held to raise public awareness of the importance of space for Spain and of strengthening our response capacity to threats to space security. Raising awareness about the importance of the space sector is part of the effort to foster a culture of national security.

The Autonomous Communities and Cities will collaborate as necessary, to the extent of their capabilities.

Training and Instruction

Aspects related to training and advanced instruction in coordinated response to crisis scenarios and potential catastrophes, and to the degradation of aerospace-related systems, will be improved by using simulation tools to prepare for such cases.

Regular inter-ministerial and international crisis management exercises will be conducted to prepare the crisis management system for different aerospace scenarios and to assess its response capability and resilience.

A Culture of Aerospace Security

The culture of space security will be reinforced in all relevant areas of public administration through regular coordination meetings, security outreach sessions, specific seminars, etc.

Aerospace safety culture will be promoted at all levels of society through a transparent, active and participatory policy of information and social communication.

Technologies that enable information sharing and interaction will be leveraged to make the communication of preventive measures and/or the consequences of risks and threats accurate, agile, consistent and coordinated.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Defence; Ministry of Home Affairs; Ministry of Science, Innovation and Universities; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency; Autonomous Communities and Cities.

MEASURES OF ACHIEVEMENT

Initiatives by stakeholders to promote space culture. Measures adopted to improve strategic communication on space security.

ACTION 13: CONTRIBUTE TO THE SPACE SECURITY ACTIVITIES OF THE INTERNATIONAL ORGANISATIONS TO WHICH SPAIN BELONGS

Spain will actively contribute to the intensification of space activities within the international initiatives of the EU Common Security and Defence Policy and within the framework of the North Atlantic Alliance. Furthermore, the inherent vulnerability and difficulty of attribution in the event of an action against satellites requires improved international cooperation in their surveillance and protection, a matter that has led to the emergence of new initiatives in this area, particularly within the European Union.

Action will be taken to promote investment and Spanish participation and representation in all international organisations, committees, programmes, forums, initiatives and working groups relevant to space security.

Action will be taken to ensure interoperability with the aerospace command and control systems of NATO/EU countries for the proper performance of their security and defence functions.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Foreign Affairs, European Union and Cooperation; Ministry of Defence; Ministry for Digital Transformation and Civil Service; Spanish Space Agency.

MEASURES OF ACHIEVEMENT

Participation in space activities and degree of contribution to the international effort in space security, particularly within the operations and missions of the European Union and NATO.

ACTION 14: STRENGTHEN COOPERATIVE SECURITY IN BILATERAL AGREEMENTS OF STRATEGIC INTEREST TO SPAIN

The Spanish Space Agency, the Ministry of Defence and public administrations with responsibilities in the space domain will carry out capacity-building activities in other countries in this respect. The purposes of these activities may be complementary. Coordinating the conception and planning of these activities under the principle of unity of action or direction is fundamental to achieving greater efficiency in the use of resources and thus more effective external action in this area.

Stronger mechanisms will be implemented for exchanging information on space surveillance with centres and agencies (civilian and military) in other nations whose coverage complements and completes that of our space surveillance system.

Spain will conclude bilateral space security agreements, primarily with neighbouring countries, and multilateral agreements, preferably with influential European countries and global space powers, for mutual support in crisis situations.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Foreign Affairs, European Union and Cooperation; Ministry of Defence; Spanish Space Agency.

MEASURES OF ACHIEVEMENT

The development of collective space security initiatives with other countries, both within international organisations and bilaterally.

ACTION 15: CONTRIBUTE TO THE SUSTAINABLE USE OF SPACE ORBITS

Spain's institutions and agencies will foster and promote the design and use of innovative technologies that enable compliance with current regulations, seeking in particular:

- To design satellites that can be safely and efficiently disposed of at the end of their useful life.
- To contribute to tracking and removing space debris from the most saturated orbits.
- To develop the capability for controlled re-entry of satellites at the end of their useful life, avoiding the creation of new space debris.
- To reduce deorbit times by using passive measures to decelerate satellites.
- To ensure satellites comply with post-mission disposal guidelines.
- To mitigate the impact of satellites on astronomy, enabling cleaner observations from Earth and helping to maintain less visually congested orbits.

Furthermore, through appropriate forums, we will collaborate with other space agencies and international organisations to establish standards and regulations to address the problem of space debris.

We will promote awareness of the importance of maintaining clean space orbits and of the need to adopt sustainable practices in space. To this end, policies and regulations will be established that include measures to ensure the sustainability of space activities.

We will participate in relevant supranational initiatives and forums aimed at sustainability in space.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Foreign Affairs, European Union and Cooperation; Ministry for the Ecological Transition and the Demographic Challenge; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency.

MEASURES OF ACHIEVEMENT

Preventive plans initiated for monitoring space debris.

Programmes that include measures to promote sustainability in space.

ACTION 16: PROMOTE THE DEVELOPMENT OF INTERNATIONAL PROTOCOLS FOR THE USE AND TRANSMISSION OF SPACE ALERTS

Action will be taken to enhance the capabilities of the Spanish Space Surveillance and Tracking System, in coordination with activities carried out within the framework of the European Space Agency, the European Union, and other areas of Spanish public administrations, to include improvements and new functionalities with respect to risk analysis and the evaluation of security measures. Specifically, we will address the following areas:

- Processing capabilities, including cataloguing and capabilities specific to space security needs;
- The capabilities to provide space surveillance and tracking services enabling more accurate calculations of object re-entry trajectories, and to detect and, where appropriate, attribute potentially dangerous manoeuvres;
- The coordination and exchange of information with other space security centres or services (in areas such as secure communications, secure navigation and remote sensing for security);
- The development and integration of national and international protocols for alerts and response to adverse space weather events, and their alignment with existing studies and plans for space weather protection.

MINISTRIES AND OTHER BODIES INVOLVED

Ministry of Foreign Affairs, European Union and Cooperation; Ministry of Defence; Ministry of Home Affairs; Ministry of Transport and Sustainable Mobility; Ministry for the Ecological Transition and the Demographic Challenge; Ministry for Digital Transformation and the Civil Service; Spanish Space Agency.

MEASURES OF ACHIEVEMENT

Protocols established for alerts and records of actions taken and resolution times.

TERMINOLOGY AND ACRONYMS

A

AEE	(Spanish) Spanish Space Agency
AEMET	(Spanish) Meteorological Agency
AENA	(Spanish) Public company responsible for airports and air navigation in Spain
AESA	(Spanish) Aviation Safety Agency
ASAT	Anti-Satellite Weapon
ASPA	Air Security and Protection Teams
ATM	Air Traffic Management

C

CBR	Chemical, Biological and Radiological
CER	Critical Entities Resiliencie (Directiva (UE) 2022/2557)
CNPIC	(Spanish) National Centre for the Protection of Critical Infrastructure

CNSA	(Spanish) National Aerospace Security Council
CNS	Communication, Navigation and Surveillance
CNSA	(Spanish) National Aerospace Security Council
COPUOS	Committee on the Peaceful Uses of Outer Space
COVE	(Spanish) Space Surveillance Operations Centre
CSN	(Spanish) National Security Council

D

DSN	(Spanish) Department of national Security

E

EA	(Spanish) Air and Space Force
EASA	European Aviation Safety Agency
EDA	European Defence Agency
EDF	European Defence Fund

EGNOS	European Geostationary Navigation Overlay Service
ENAIRE	(Spanish) Air Navigation Manager
EOGS	Earth Observation Governmental Service
ENS	National Security Scheme (R.D. 311/2022, from May 3)
ESA	European Space Agency
ESAN	National Aerospace Security Strategy
ESN	National Security Strategy
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
EU SSSD	EU Space Strategy for Security and Defence
EU SST	EU Space Surveillance and Tracking
EUSPA	EU Agency for the Space Programme
eVTOL	Electric Vertical Take-Off and Landing

F

FAS	(Spanish) Armed Forces

G

GNSS	Global Navigation Satellite System
GOVSATCOM	Governmental Satellite Communications

H

HAO	Higher Airspace Operations
HAPS	High-Altitude Pseudo-Satellites

I

IA	Inteligencia Artificial
IADC	Inter-Agency Space Debris Coordination Committee
IAMD	Integrated Air and Missile Defence
ICAO	International Civil Aviation Organisation
ICARO	(Spanish) Identification of Anomalous Conduct and Operational Response
INTA	National Institute of Aerospace Technology
IOS	In-Orbit Servicing

IRIS2	Infrastructure for Resilience, Interconnectivity and Security by Satellite
ISAC	Spatial Information Exchange Center
J	
JEMAD	(Spanish) Chief of Defence
L	
LEO	Low Earth Orbit
M	
MDDU	Defense Material and Dual Use Products and Technologies
MESPA	(Spanish) Space Command
MITMS	(Spanish) Ministry of Transport and Sustainable Mobility
MTOM	Maximun Take-off Mass
N	
NASS	National Aerospace Security Strategy

NEO	Near Earth Objects
NEI	Non-Earth Imaging
NIS2	Network and Information Systems (Directiva (UE) 2022/2555)
NSS	National Security Strategy

P

PART-IS	Reglamento (UE) 2023/203 y Reglamento Delegado (UE) 2022/1645
PESC	Common Foreign and Security Policy of the European Union
PERTE	(Spanish) Strategic Project for Aerospace Economic Recovery and Transformation
PNT	Posicionamiento, Navegación y Tiempo
PTA	(Spanish) Aeronautical Technology Programme
PTE	(Spanish) Space Technology Programme

Q

QRA	Quick Reaction Alert

R

R&D	Research & Development
RPOs	Rendezvous and Proximity Operations
RSP	Recognized Space Picture

S

S3TOC	(Spanish) Space Surveillance and Tracking Operations Centre
S3TSR	Spanish Space Surveillance and Tracking Surveillance Radar
SAF	Sustainable Aviation Fuel
SAR	Synthetic Aperture Radar
SDA	Space Domain Awareness
DSGs	Sustainable Development Goals
SIAC	Single Intelligence Analysis Capacity
SIGLO-CD	Global Counter Drone System
SME	Small and Medium-sized Enterprises
SSA	Space Situational Awareness
SST	Space Surveillance and Tracking

STRA	Space Threat Response Arquitecture
SVICA	(Spanish) Aerospace Surveillance and Control System
SWE	Space Weather Events

T

TLR	Technological Readiness Level

U

UAS	Unmanned Aircraft Systems
UNCLOS	United Nations Convention on the Law of the Sea
UNESEV	(Spanish) National Flight Security Escort Units
UNOOSA	United Nations Office for Outer Space Affairs

V

VLEO	Very Low Earth Orbit